

BACHELOR THESIS

Comparison of Survival Prediction Models with LASSO and Random Survival Forests for Patients with Liver Disease

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1 Introduction

The human liver is a multifunctional organ essential for metabolic regulation, detoxification, protein synthesis, and immune surveillance. These functions are coordinated by hepatocytes and various non-parenchymal cells. Chronic liver disease disrupts these intricate cellular networks, resulting in both structural and functional impairment, which impairs the quality of daily life. This progression contributes significantly to global morbidity and mortality. Because chronic liver diseases often share overlapping biochemical and diagnostic features, assessing disease severity and prognosis remains challenging (Gan et al., 2025, p.1). Reliable survival prediction is therefore crucial, as it supports risk classification, therapeutic decision-making, and long-term patient management. Survival analysis provides a statistical framework for modeling time-to-event outcomes, accounting for censored observations and patient-specific covariates.

The increasing availability of clinical data, including Ramayan Spectroscopy features, has enabled survival analysis to extend beyond traditional low-dimensional regression settings such as the Cox proportional hazards framework (Cox, 1972), which remains the standard due to its interpretability and well-established assumptions. The LASSO-penalized Cox model (Friedman et al, 2010), a regularized extension, facilitates variable selection and coefficient shrinkage, making it suitable for high-dimensional data. Random Survival Forests (Ishwaran et al, 2008) offer an alternative that captures nonlinear effects and interactions without relying on the proportional hazards assumption. Hybrid and ensemble-based survival modeling strategies seek to integrate the strengths of structured regression models with the flexibility of more advanced survival estimators.

This thesis aims to compare survival prediction models for patients with liver disease, specifically evaluating the Cox proportional hazards model with LASSO regularization, Random Survival Forests, and two combined strategies that integrate these established methods. The first combined approach is the ensemble-based Tree-Cluster Cox method, proposed by Shamsutdinova et al. (2022), which aggregates predictions from multiple survival models to enhance predictive accuracy and relax the proportional hazards assumption. The second is a hybrid approach that utilizes Random Survival Forest-based variable importance measures as a preliminary step prior to subsequent survival modeling.

To achieve this objective, multiple predefined predictor sets are evaluated using a nested cross-validation framework, which maintains unbiased model assessment. Model perfor-

mance is assessed with time-dependent survival metrics, including Integrated Prediction Accuracy (IPA) and the Brier score, enabling systematic comparison of individual and combined modeling approaches.

The results indicate that survival prediction performance differs depending on modeling approaches and predictor combinations, especially the application of the hybrid approach method shows improvement in prediction accuracy compared to single-model approaches like Cox model. This indicates the benefits of integrating another complementary modeling methods together.

This thesis first introduces the dataset, the survival outcome, and the construction of various predictor combinations. It then details the statistical methods employed, including the Cox model, LASSO regularization, Random Survival Forests, the hybrid approach, and the Tree-Cluster Cox method. The nested cross-validation strategy and relevant performance metrics such as IPA and the Brier score are also discussed. The results section analyzes model performance across different variable combinations. The thesis concludes by summarizing the findings, addressing limitations, and suggesting possible improvements for future research.

2 Problem Statement

In this chapter, the data used this thesis are described in detail. First, the background and data collection process are introduced. This is followed by a combined description of the data material, included variables, and data quality aspects. Finally, the objectives of the analysis are presented.

2.1 Data Background and Data Collection

The data used in this thesis originate from a clinical observational study of patients diagnosed with liver disease. It is provided by Prof. Dr. Jan G. Hengstler from Leibniz Research Centre for Working Environment and Human Factors, Dortmund. The dataset consists of routinely collected clinical information obtained during hospital treatment and follow-up.

Patient information was recorded at baseline and during subsequent follow-up periods. Censoring occurs when a patient has not experienced the event by the end of the observation period or is lost to follow-up.

Outcome definition. The outcome of interest in this thesis is all-cause mortality. Time-to-event is measured in months using the variable `TFS_months`. Event is derived from the variable `TFS_Event_012`, where death (2) is treated as the event of interest. Observations with liver transplantation (1) are treated as right-censored at the time of transplantation, and patients without death or transplantation during follow-up are censored at the end of observation.

All analysis conducted in this thesis are therefore based on the framework with death as the event of interest.

Predictor variables are grouped into five main categories.

- **Clinical characteristics** include demographic information (age, sex), anthropometric measures (body mass index), indicators of chronic liver disease, and established disease severity scores such as the Child–Pugh score and the Model for End-Stage Liver Disease (MELD). In addition, clinical events and follow-up indicators (e.g. hepatic encephalopathy, acute kidney injury, infections) are recorded together with their corresponding time-to-occurrence variables in months.
- **Laboratory measurements** comprise routinely collected blood-based parameters reflecting liver function, kidney function, hematological status, and metabolic balance, including electrolytes, liver enzymes, coagulation parameters, and serum proteins.
- **Bile acid profile (absolute measures)** consist of quantitative measurements of individual bile acid species obtained from targeted bile acid profiling. These variables represent absolute concentrations and are recorded on a continuous scale.
- **Raman spectroscopy features (RAMA variables)** comprise a high-dimensional set of continuous variables derived from Raman spectroscopy, indexed from `x202` to `x3148`.

Each variable represents the intensity at a specific Raman shift. All RAMA variables are fully observed and contain no missing values.

Additional variables present in the dataset but not used in the subsequent analyses are documented in the data dictionary but are not discussed in detail.

2.2 Data Material and Data Quality

The dataset comprises a group of patients with chronic liver disease, where each observation corresponds to a single patient.

Variables are measured on multiple scales, including continuous, binary, and categorical levels. Continuous variables are recorded in standard clinical units, while categorical variables follow clinically meaningful groupings. Composite scores are interpreted according to established clinical definitions. These scores summarize multiple clinical and laboratory parameters into a single index, with higher values indicating more advanced disease severity (e.g. Child–Pugh and MELD scores).

Missing values are present in several predictor variables. Tables 1, 2, and 3 summarize all variables with non-zero missingness, clasified by variable type. Variables without missing values are omitted from these tables. The complete description of all variables, including definitions, coding schemes, and units of measurement, is provided in Appendix A.

Table 1: Clinical characteristics with missing values.

Variable	Missing (n)	Missing (%)
HVPG (3-group stratification, 16 mmHg)	193	42.23
Baseline liver stiffness	68	14.88
Varices (binary)	5	1.09
Chronic kidney disease (CNI_01)	1	0.22
Baseline diabetes mellitus (bl_dm_012)	2	0.44
Baseline body mass index (BMI)	1	0.22
ICG plasma disappearance rate (ICG-PDR)	234	51.20
PCG-R15	238	52.08

Missingness among laboratory variables, shown in Table 2, is generally low for routine blood parameters but more prominent for specialized biomarkers, particularly immunoglobulin subclasses and fibrosis-related markers, reflecting selective availability in clinical practice.

Table 2: Laboratory measurements with missing values.

Variable	Missing (n)	Missing (%)
Hemoglobin (hb)	2	0.44
Platelet count (plt)	4	0.88
White blood cell count (wbc)	2	0.44
Potassium (k)	42	9.19
Chloride (cl)	37	8.10
Blood urea nitrogen (bun)	3	0.66
Total protein	3	0.66
Cholinesterase (che)	2	0.44
Alkaline phosphatase (ap)	3	0.66
Aspartate aminotransferase (ASAT)	3	0.66
Alanine aminotransferase (ALAT)	1	0.22
Gamma-glutamyl transferase (GGT)	1	0.22
Lactate dehydrogenase (LDH)	7	1.53
Amylase (aAMY)	40	8.75
Pancreatic amylase (pAMY)	58	12.69
Lipase (lip)	40	8.75
Fibrinogen (fgen)	1	0.22
Bile acids (BA)	8	1.75
Glucose (gluc)	8	1.75
High-density lipoprotein (HDL)	2	0.44
Low-density lipoprotein (LDL)	4	0.88
Glycated hemoglobin (HbA1c)	66	14.44
Ammonia (NH ₃)	7	1.53
C-reactive protein (CRP)	1	0.22
Lipopolysaccharide-binding protein (LBP)	3	0.66
Immunoglobulin A (IgA)	101	22.10
Immunoglobulin G (IgG)	139	30.42
Immunoglobulin M (IgM)	142	31.07
IgG subclass 1	142	31.07
IgG subclass 2	142	31.07
IgG subclass 3	142	31.07
IgG subclass 4	142	31.07
B-type natriuretic peptide (BNP)	3	0.66
TIMP-1	2	0.44
Procollagen III N-terminal peptide (P3NP)	2	0.44
Hyaluronic acid (HA)	3	0.66

Table 3: Bile acid profile variables (absolute measures) with missing values.

Variable	Missing (n)	Missing (%)
TMCA	137	29.98
TCA sulfate	328	71.77
TUDCA	73	15.97
TCA	2	0.44
oMCA	150	32.82
GUDCA	7	1.53
CA sulfate	452	98.91
aMCA	96	21.01
bMCA	436	95.40
TDCA	40	8.75
CA	8	1.75
UDCA	66	14.44
GDCA	10	2.19
TLCA	187	40.92
CDCA	2	0.44
DCA	53	11.60
LCA	77	16.85
C4	42	9.19

2.3 Construction of Predictor Combinations

To assess the contribution of different predictor variables to survival prediction, predictor variables were grouped into four main domains: laboratory measurements (L), bile-acid profile variables (B), Raman spectroscopy features (R), and clinical characteristics (C). Based on these domains, predefined predictor combinations were constructed to evaluate both isolated and joint effects of the different predictor types combinations. All non-empty combinations of these four domains were considered, resulting in a total of $2^4 - 1 = 15$ predictor-group combinations.

Specifically, the evaluated combinations were:

- Single-domain models: lab(L), bile acids(B), raman(R), clinical(C)
- Two-domains models: lab+ba(L+B), lab+raman(L+R), lab+clinical(L+C), ba+raman(B+R), ba+clinical(B+C), raman+clinical(R+C)
- Three-domains models: lab+ba+raman(L+B+R), lab+ba+clinical(L+B+C), lab+raman+clinical(L+R+C), ba+raman+clinical(B+R+C)
- Four-domains model: lab+ba+raman+clinical(L+B+R+C)

These combinations were defined beforehand and were not selected based on any prior observed performance. This allows for structured comparison of how different biological and clinical information sources contribute individually and in combination to predictive accuracy. By evaluating all possible domains combinations, it allows us identify whether additional predictor domains provide further prediction value beyond established clinical and laboratory variables.

2.4 Objective of the Project

The objective of this project is to compare different survival prediction models for patients with liver disease with respect to their predictive performance and robustness. Several survival modeling strategies are compared, including a Cox proportional hazards model with LASSO regularization, Random Survival Forests, a hybrid modeling approach, and an ensemble-based method (Tree-Cluster Cox).

Model performances are then evaluated using time-dependent survival metrics, namely the Integrated Prediction Accuracy (IPA) and the Brier score. A nested cross-validation framework is used to obtain unbiased performance estimates and to assess model stability across multiple predefined predictor combinations. The main goal is to determine whether combined modeling strategies provide advantages over single-model approaches in the context of survival prediction for patients with liver disease.

3 Statistical Methods

This section explains the various statistical methods used in this project, together with the software used.

3.1 Cox Proportional Hazards Model

Survival analysis aims to model the time until the occurrence of an event of interest. The Cox proportional hazards model, introduced by Cox (1972, p. 187–201), is one of the most widely used regression models for time-to-event data. Let T be a non-negative random variable representing the event time. The survival function is defined as

$$S(t) = \mathbb{P}(T > t),$$

and describes the probability that the event has not occurred by time t . Closely related is the hazard function

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbb{P}(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t},$$

which represents the instantaneous event rate at time t conditional on survival up to that time. In regression-based survival modeling, the objective is to describe how the hazard function depends on a vector of covariates

$$\mathbf{Z} = (Z_1, \dots, Z_p)^\top.$$

The Cox proportional hazards model specifies the conditional hazard function as

$$\lambda(t \mid \mathbf{Z}) = \lambda_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{Z}) = \lambda_0(t) \exp\left(\sum_{k=1}^p \beta_k Z_k\right),$$

where $\lambda_0(t)$ is the unspecified baseline hazard function and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$ is a vector of regression parameters. Proportional hazards assumption is a feature of the Cox Model. For two individuals with covariate vectors $\mathbf{Z}$ and $\tilde{\mathbf{Z}}$, the hazard ratio is given by

$$\frac{\lambda(t \mid \mathbf{Z})}{\lambda(t \mid \tilde{\mathbf{Z}})} = \exp(\boldsymbol{\beta}^\top (\mathbf{Z} - \tilde{\mathbf{Z}})),$$

which does not depend on time t . Thus, the ratio of the hazard functions for two individuals remains constant over time. Each regression coefficient β_k quantifies the effect of the corresponding covariate Z_k on the hazard. A one-unit increase in Z_k , holding all other covariates constant, multiplies the hazard by $\exp(\beta_k)$. Values $\exp(\beta_k) > 1$ indicate an increased risk of the event, whereas values $\exp(\beta_k) < 1$ indicate a decreased risk. A key advantage of the Cox proportional hazards model is its semi-parametric nature. The regression parameters $\boldsymbol{\beta}$ can be estimated without specifying the baseline hazard function $\lambda_0(t)$. Estimation of the regression parameter vector $\boldsymbol{\beta}$ is performed by maximizing the partial likelihood introduced by Cox. Let

$$t_1 < t_2 < \dots < t_D$$

denote the ordered observed event times. For each event time t_i , let

$$\mathbf{Z}_i = (Z_{i1}, \dots, Z_{ip})^\top$$

denote the covariate vector of the individual experiencing the event at time t_i , and let

$$R(t_i) = \{j : T_j \geq t_i\}$$

be the corresponding risk set, i.e. the set of all individuals who are still under risk at time t_i . Under the Cox proportional hazards model

$$\lambda(t \mid \mathbf{Z}) = \lambda_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{Z}),$$

the contribution of the event occurring at time t_i to the partial likelihood is given by

$$L_i(\boldsymbol{\beta}) = \frac{\exp(\boldsymbol{\beta}^\top \mathbf{Z}_i)}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}^\top \mathbf{Z}_j)}.$$

This quantity can be interpreted as the conditional probability that, among all individuals in the risk set $R(t_i)$, the individual with covariates $\mathbf{Z}_i$ experiences the event at time t_i . The partial likelihood is defined as the product of these contributions over all observed event times,

$$L(\boldsymbol{\beta}) = \prod_{i=1}^D L_i(\boldsymbol{\beta}) = \prod_{i=1}^D \frac{\exp(\boldsymbol{\beta}^\top \mathbf{Z}_i)}{\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}^\top \mathbf{Z}_j)}.$$

Importantly, the baseline hazard function $\lambda_0(t)$ cancels out and does not appear in the partial likelihood. Consequently, the regression parameters $\boldsymbol{\beta}$ can be estimated without specifying $\lambda_0(t)$. Estimation of $\boldsymbol{\beta}$ is carried out by maximizing the corresponding log-partial likelihood,

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^D \boldsymbol{\beta}^\top \mathbf{Z}_i - \sum_{i=1}^D \log \left(\sum_{j \in R(t_i)} \exp(\boldsymbol{\beta}^\top \mathbf{Z}_j) \right),$$

which is solved numerically. The resulting maximum partial likelihood estimator $\hat{\boldsymbol{\beta}}$ provides consistent estimates of the covariate effects under the proportional hazards assumption. After estimation of the regression parameters, the baseline hazard and the corresponding baseline survival function can be estimated if required. The conditional survival function for an individual with covariates $\mathbf{Z}$ is given by

$$S(t \mid \mathbf{Z}) = S_0(t)^{\exp(\boldsymbol{\beta}^\top \mathbf{Z})},$$

where $S_0(t)$ denotes the baseline survival function.

Estimation of the baseline hazard (Breslow estimator). While the regression parameter β is estimated via maximization of the partial likelihood, the cumulative baseline hazard function

$$\Lambda_0(t) = \int_0^t \lambda_0(u) du$$

can be obtained within the joint likelihood framework. Following Lin (2007, pg 471-477), consider the full likelihood

$$L(\beta, \Lambda_0) = \prod_{i=1}^n \left\{ \exp(\beta^\top \mathbf{Z}_i(T_i)) \lambda_0(T_i) \right\}^{\Delta_i} \exp \left(- \int_0^{T_i} \exp(\beta^\top \mathbf{Z}_i(u)) \lambda_0(u) du \right),$$

where Δ_i indicates event occurrence. For each subject $i = 1, \dots, n$, the observed follow-up time is $T_i = \min(\tilde{T}_i, C_i)$, where $\tilde{T}_i$ is the true event time and C_i is the censoring time. The vector $\mathbf{Z}_i(t) = (Z_{i1}(t), \dots, Z_{ip}(t))^\top$ denotes the p -dimensional covariate vector of subject i at time t .

By treating $\lambda_0(t)$ as piecewise constant between distinct uncensored event times, it can be shown that the joint likelihood is maximized simultaneously at the maximum partial likelihood estimator $\hat{\beta}$ and at the estimator

$$\hat{\Lambda}_0(t) = \sum_{i=1}^n \frac{I(T_i \leq t) \Delta_i}{\sum_{j \in R(T_i)} \exp(\hat{\beta}^\top \mathbf{Z}_j(T_i))},$$

which is known as the Breslow estimator of the cumulative baseline hazard. The indicator function $I(T_i \leq t)$ equals 1 if $T_i \leq t$ and 0 otherwise. Thus, the numerator counts the number of observed events up to time t , while the denominator represents the total weighted risk at each event time.

In counting-process notation, this estimator can equivalently be written as

$$\hat{\Lambda}_0(t) = \sum_{i=1}^n \int_0^t \frac{dN_i(u)}{\sum_{j=1}^n Y_j(u) \exp(\hat{\beta}^\top \mathbf{Z}_j(u))},$$

where $N_i(t)$ records the number of events observed on the i -th subject by time t , and $Y_i(t)$ indicates whether the i th subject is at risk at time t , and $\mathbf{Z}_i(t)$ is the set of covariates at time t .

The corresponding estimator of the conditional survival function for an individual with covariate vector $\mathbf{Z} = z$ is

$$\hat{S}(t | z) = \exp\left(-\int_0^t \exp\left(\hat{\boldsymbol{\beta}}^\top z(u)\right) d\hat{\Lambda}_0(u)\right).$$

3.2 LASSO-Regularized Cox Proportional Hazards Model

In high-dimensional settings or in the presence of correlated covariates, classical Cox regression may lead to unstable estimates and poor predictive performance. To address these issues, regularization techniques can be incorporated into the Cox proportional hazards model. In this thesis, LASSO regularization is applied using the `glmnet` package in R, following the framework of Friedman et al. (2010, p. 2-11).

Let $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$ denote the regression parameter vector in the Cox proportional hazards model

$$\lambda(t | \mathbf{Z}) = \lambda_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{Z}).$$

As described in the previous subsection, estimation in the unpenalized Cox model is based on maximizing the partial log-likelihood $\ell(\boldsymbol{\beta})$.

Penalized objective function. The LASSO-regularized Cox model maximizes a penalized partial log-likelihood of the form

$$\hat{\boldsymbol{\beta}}(\lambda, \alpha) = \arg \max_{\boldsymbol{\beta}} \{\ell(\boldsymbol{\beta}) - \lambda P_\alpha(\boldsymbol{\beta})\},$$

where $\lambda \geq 0$ is a tuning parameter controlling the overall strength of regularization and $P_\alpha(\boldsymbol{\beta})$ denotes the elastic-net penalty

$$P_\alpha(\boldsymbol{\beta}) = \sum_{k=1}^p \left(\frac{1-\alpha}{2} \beta_k^2 + \alpha |\beta_k| \right), \quad 0 \leq \alpha \leq 1.$$

The parameter α controls the type of regularization. Setting $\alpha = 1$ corresponds to LASSO regularization, while $\alpha = 0$ yields ridge regression. Here, $\alpha = 1$ is used, resulting in an ℓ_1 penalty that forces many regression coefficients to become exactly zero.

Coordinate descent optimization. The optimization problem is convex and is solved in `glmnet` using cyclical coordinate descent. At each iteration, the objective function

is optimized with respect to a single parameter β_k while keeping all other parameters fixed.

For standardized covariates, the coordinate-wise update for β_k takes the form

$$\beta_k \leftarrow \frac{S(u_k, \lambda\alpha)}{1 + \lambda(1 - \alpha)},$$

where $S(\cdot, \cdot)$ denotes the soft-thresholding operator

$$S(z, \gamma) = \begin{cases} z - \gamma, & z > \gamma, \\ z + \gamma, & z < -\gamma, \\ 0, & |z| \leq \gamma, \end{cases}$$

and u_k represents the partial derivative of the partial log-likelihood with respect to β_k , evaluated at the current parameter values. The soft-thresholding step is responsible for shrinking small coefficients exactly to zero, thereby performing variable selection.

Pathwise optimization and warm starts. Rather than solving the penalized optimization problem for a single value of λ , `glmnet` computes solutions along a decreasing sequence of λ values,

$$\lambda_{\max} > \lambda_1 > \cdots > \lambda_L,$$

where $\lambda_{\max}$ is chosen such that all regression coefficients are zero. The solution for a given λ_ℓ is used as the starting point (warm start) for optimization at $\lambda_{\ell+1}$, which greatly improves computational efficiency and stability. This procedure yields an entire regularization path of models with increasing complexity as λ decreases.

Standardization and intercept handling. Prior to model fitting, all covariates are standardized to have mean zero and unit variance. This ensures that the penalty is applied uniformly across covariates. The intercept term is not penalized and is treated separately from the regression coefficients.

Selection of the tuning parameter. The tuning parameter λ is selected using K -fold cross-validation. For each value of λ along the regularization path, the model is fitted on the training folds and evaluated on the held-out fold using an appropriate prediction error criterion. The value $\lambda_{\min}$ corresponding to the minimum cross-validated error is

commonly used, while a more parsimonious alternative is given by the largest λ whose error is within one standard error of the minimum (the one-standard-error rule).

Interpretation. The resulting LASSO-penalized Cox model yields sparse regression coefficients, where only a subset of covariates has non-zero effects on the hazard. The estimated coefficients are interpreted analogously to the unpenalized Cox model via hazard ratios $\exp(\beta_k)$, but only for variables retained in the model. This combination of regularization and interpretability makes the LASSO-penalized Cox model particularly suitable for high-dimensional clinical survival data.

3.3 Log-rank Test

The log-rank test (Mantel, 1966, p. 163-170) is a nonparametric statistical test used to compare survival distributions between two or more groups in the presence of right-censored data. It is commonly applied in conjunction with Kaplan–Meier estimators to assess whether observed differences between group-specific survival curves are statistically significant.

Let T denote the survival time and δ the event indicator, where $\delta = 1$ indicates occurrence of the event of interest and $\delta = 0$ denotes right censoring. Suppose the sample is partitioned into G groups based on a categorical covariate. At each distinct event time t_j , let d_{gj} denote the number of observed events in group g at time t_j , and let Y_{gj} denote the number of individuals at risk in group g just prior to t_j . Furthermore, let $d_j = \sum_{g=1}^G d_{gj}$ and $Y_j = \sum_{g=1}^G Y_{gj}$ denote the total number of events and individuals at risk at time t_j , respectively.

Under the null hypothesis

$$H_0 : S_1(t) = S_2(t) = \cdots = S_G(t) \quad \text{for all } t,$$

the survival functions are identical across all comparison groups. Here, G denotes the number of groups defined by a categorical covariate, and $S_g(t)$ represents the survival function of group g , defined as

$$S_g(t) = \Pr(T > t \mid \text{group } g),$$

that is, the probability that the event time T exceeds time t for an individual belonging to group g . The equality of survival functions implies that individuals in all groups share the same time-to-event distribution.

The alternative hypothesis states that at least one group has a survival function that differs from the others, indicating a difference in survival experience between groups.

Under H_0 , the expected number of events in group g at time t_j is given by

$$E_{gj} = \frac{Y_{gj}}{Y_j} d_j.$$

The log-rank test statistic compares the observed and expected number of events accumulated over all event times,

$$Z_g = \sum_j (d_{gj} - E_{gj}).$$

For the two-group case, the test statistic can be written as

$$\chi^2 = \frac{Z_1^2}{\text{Var}(Z_1)},$$

which asymptotically follows a chi-squared distribution with one degree of freedom under the null hypothesis. For comparisons involving more than two groups, the test statistic follows a chi-squared distribution with $G - 1$ degrees of freedom. The decision rule of the test is based on a pre-specified significance level α . The null hypothesis H_0 is rejected if the observed test statistic exceeds the $(1 - \alpha)$ -quantile of the corresponding chi-squared distribution, or equivalently if the associated p -value is smaller than α . Otherwise, H_0 is not rejected.

3.4 Random Survival Forests

Random Survival Forests (RSF) are an ensemble tree method for survival analysis proposed by Ishwaran et al. (2008, p. 841-860). RSF extends Breiman's random forests to right-censored time-to-event data by constructing an ensemble of survival trees, where splitting rules and terminal node predictions explicitly incorporate survival information. RSF is particularly useful when covariate effects may be nonlinear or involve complex interactions, since such structures can be captured automatically by tree-based models.

Notation. Let T denote the event time and $\mathbf{Z} = (Z_1, \dots, Z_p)^\top$ the covariate vector. A survival tree partitions the covariate space into terminal nodes (leaves). All individuals assigned to the same terminal node share a common estimated survival quantity such as a cumulative hazard function.

RSF Algorithm

Given a dataset of n individuals, RSF constructs an ensemble of B survival trees using the following steps:

1. **Bootstrap sampling:** For each tree $b = 1, \dots, B$, draw a bootstrap sample of size n from the original data (sampling with replacement). Observations not selected for the bootstrap sample form the *out-of-bag* (OOB) data for tree b .
2. **Grow a survival tree:** Using the bootstrap sample, grow a binary survival tree by recursive splitting. At each node, randomly select m_{try} candidate covariates from the p available covariates. Among these candidates, choose the split that maximizes a survival-based splitting criterion (e.g. log-rank). Repeat until a stopping rule is met.
3. **Terminal node constraint:** Trees are typically grown to full size, subject to a minimum number of unique events (deaths) in each terminal node. Denote this minimum by $d_0 > 0$; splitting stops when a node contains fewer than d_0 unique events.
4. **Node-wise survival estimation:** For each terminal node, estimate the cumulative hazard function (CHF) based on individuals falling into that node, and assign this CHF to any new observation that ends up in the same node.
5. **Ensemble prediction:** Aggregate predictions across trees to obtain an ensemble CHF (or related survival quantity).
6. **OOB error estimation:** Use OOB observations to estimate prediction error without an explicit external validation split.

Survival Tree Construction and Log-rank Splitting

A survival tree is a binary tree grown by recursively partitioning the covariate space into subgroups with distinct survival distributions. At a given node h , all observations

belonging to that node are considered, and candidate splits are evaluated for each covariate.

For a continuous covariate Z_k , candidate splits are of the form $Z_k \leq c$ versus $Z_k > c$, where c ranges over all admissible split points that respect a minimum node size constraint. For categorical covariates, splits are defined by partitioning the set of categories into two non-empty subsets.

To evaluate a candidate split, the resulting left and right daughter nodes are compared using a two-sample log-rank statistic. Let $Y_1(t)$ and $Y_2(t)$ denote the numbers at risk in the two daughter nodes at time t , and let $N_1(t)$ and $N_2(t)$ denote the corresponding numbers of observed events. Following LeBlanc and Crowley, (1993, p. 457-465), the log-rank splitting statistic can be expressed as a weighted difference between the estimated hazard increments of the two groups,

$$G = \int w(t) \frac{Y_1(t)Y_2(t)}{Y_1(t) + Y_2(t)} (d\hat{\Lambda}_1(t) - d\hat{\Lambda}_2(t)),$$

where $w(t) = 1$ corresponds to the standard log-rank test and $\hat{\Lambda}_j(t)$ denotes the Nelson–Aalen cumulative hazard estimator for group j .

In practice, the squared log-rank statistic standardized by an estimate of its variance is used as the splitting criterion. For a given node h , the optimal split s^* is defined as

$$s^* = \arg \max_{s \in \mathcal{S}_h} G(s, h),$$

where $\mathcal{S}_h$ denotes the set of all admissible splits at node h . The selected split is therefore the one that maximizes the separation between survival distributions in the two daughter nodes.

This splitting procedure is applied recursively to each newly created node until a stopping criterion is reached, such as a minimum node size.

The use of the log-rank statistic as a splitting rule is particularly attractive because it is well suited for right-censored survival data and has been shown to be stable under varying degrees of censoring (LeBlanc and Crowley, 1993, p. 457-465). Moreover, the log-rank statistic can be interpreted as a standardized measure of dissimilarity between empirical hazard functions, making it a natural criterion for constructing prognostic risk groups in survival trees.

Terminal Node Prediction via Nelson–Aalen CHF

Let h denote a terminal node and let the distinct event times observed among individuals in node h be $t_{1,h} < t_{2,h} < \dots < t_{K_h,h}$. Let $d_{l,h}$ be the number of events at time $t_{l,h}$ within node h , and let $Y_{l,h}$ be the number of individuals at risk just prior to $t_{l,h}$ within node h . The terminal-node cumulative hazard function is estimated using the Nelson–Aalen estimator:

$$\widehat{H}_h(t) = \sum_{t_{l,h} \leq t} \frac{d_{l,h}}{Y_{l,h}}.$$

If an individual with covariates $\mathbf{Z}$ falls into terminal node h , the tree-predicted CHF is defined as

$$\widehat{H}(t \mid \mathbf{Z}) = \widehat{H}_h(t).$$

This provides a nonparametric survival estimate within each terminal node.

Ensemble CHF and OOB Ensemble CHF

Let $\widehat{H}^{(b)}(t \mid \mathbf{Z})$ denote the CHF predicted by tree b . The RSF ensemble CHF is obtained by averaging over all B trees:

$$\widehat{H}_{\text{ens}}(t \mid \mathbf{Z}) = \frac{1}{B} \sum_{b=1}^B \widehat{H}^{(b)}(t \mid \mathbf{Z}).$$

For OOB prediction, only trees for which the observation was OOB are used. Let $I_{i,b} = 1$ if observation i is OOB for tree b and $I_{i,b} = 0$ otherwise. Then the OOB ensemble CHF for observation i is

$$\widehat{H}_{\text{ens}}^{\text{OOB}}(t \mid \mathbf{Z}_i) = \frac{\sum_{b=1}^B I_{i,b} \widehat{H}^{(b)}(t \mid \mathbf{Z}_i)}{\sum_{b=1}^B I_{i,b}}.$$

OOB prediction enables internal performance assessment and is also used in the definition of variable importance.

Variable Importance (VIMP)

RSF provides a variable importance measure (VIMP) to quantify the predictive contribution of each covariate. VIMP for a variable Z_k measures the change in prediction error when the association between Z_k and the outcome is destroyed.

In the approach described by Ishwaran et al. (2008), VIMP is computed using OOB data by

1. computing the OOB prediction error of the original forest,
2. randomly permuting (or otherwise randomizing) the values of Z_k in the OOB data,
3. recomputing the OOB prediction error using the modified data,
4. defining VIMP as the increase in error due to the permutation.

Large positive VIMP values indicate that the variable is important for prediction, whereas values near zero suggest little predictive contribution. Negative values may occur due to random variation and typically indicate that the variable does not improve prediction. VIMP should be interpreted with care. It does *not* estimate the difference in prediction error between two forests refit with and without a given variable. Instead, VIMP quantifies the change in prediction error when the variable is rendered unavailable *at prediction time*, while keeping the originally fitted forest fixed. Consequently, in the presence of highly correlated predictors, multiple variables may simultaneously exhibit large VIMP values. Removing one correlated variable and refitting the forest can substantially change the VIMP of the remaining variables, even if the overall prediction error remains essentially unchanged. Thus, VIMP is best understood as a measure of predictive reliance of the fitted forest on a variable, rather than as a direct measure of the variable's unique contribution in a refitted model.

3.5 Brier Score for Survival Data

The Brier score is a proper scoring rule for evaluating probabilistic predictions and was originally introduced by Brier (1950, p. 1-3) in the context of forecast verification. In survival analysis, the Brier score is used to assess the accuracy of predicted event-free probabilities at a fixed prediction horizon and was extended to right-censored time-to-event data by Graf et al. (1999, p. 2530-2541).

Expected Brier score. Let T denote the event time and let $t^* > 0$ be a fixed prediction horizon. Let $\hat{\pi}(t^* \mid \mathbf{Z}) = \mathbb{P}(T > t^* \mid \mathbf{Z})$ denote the predicted event-free (survival) probability at time t^* given covariates $\mathbf{Z}$. The expected Brier score at time t^* is defined as

$$\mathbb{E} \left[(\mathbb{I}(T > t^*) - \hat{\pi}(t^* \mid \mathbf{Z}))^2 \right],$$

which corresponds to the mean squared error of the predicted event-free probabilities.

Empirical Brier score. Given a sample of n individuals with covariates $\mathbf{Z}_1, \dots, \mathbf{Z}_n$, the empirical Brier score at time t^* is defined as

$$\text{BS}(t^*) = \frac{1}{n} \sum_{i=1}^n (\mathbb{I}(T_i > t^*) - \hat{\pi}(t^* | \mathbf{Z}_i))^2.$$

The Brier score takes values between 0 and 1, with smaller values indicating better predictive accuracy. A score of 0 corresponds to perfect prediction.

Handling right-censoring. In the presence of right-censored observations, the indicator $\mathbb{I}(T_i > t^*)$ may not be observable for all individuals. Graf et al. (1999) proposed estimating the Brier score using inverse probability of censoring weighting (IPC weighting). This approach reweights each contribution by the inverse of the estimated censoring distribution, yielding an unbiased estimator of the Brier score under the assumption of independent censoring.

Interpretation and decomposition. Graf et al. (1999) show that the expected Brier score admits a decomposition into two components: an *imprecision* term, which reflects how well the predicted probabilities approximate the true event-free probabilities, and an *irreducible* term, which depends only on the true survival distribution. The expected Brier score is minimized when the predicted probabilities equal the true conditional event-free probabilities, establishing the Brier score as a strictly proper scoring rule.

Integrated Brier score. Rather than evaluating prediction accuracy at a single time point, prediction error can be summarized over a time interval $[0, \tau]$ by integrating the Brier score with respect to a weight function $W(t)$:

$$\text{IBS} = \frac{1}{n} \sum_{i=1}^n \int_0^\tau (\mathbb{I}(T_i > t) - \hat{\pi}(t | \mathbf{Z}_i))^2 dW(t).$$

Natural choices for $W(t)$ include weights based on the marginal survival distribution. The integrated Brier score provides a global summary of predictive accuracy over time.

3.6 Index of Prediction Accuracy (IPA)

To evaluate and compare the predictive performance of survival models, the Index of Prediction Accuracy (IPA) is used. IPA is defined as a rescaled version of the time-dependent Brier score described in Section 3.5. For survival data, the Brier score is estimated using the inverse probability of censoring weighted estimator proposed by Graf et al. (1999), and IPA rescales this quantity relative to a null model as proposed by Kattan and Gerds (2018, p. 2-6).

Brier score for survival prediction. Let T denote the event time and let $t > 0$ be a fixed prediction horizon. For an individual i , let $\hat{\pi}(t^* \mid \mathbf{Z})$ denote the model-based predicted survival probability at time t . The Brier score at time t is defined as

$$\text{BS}(t) = \mathbb{E} \left[(\mathbb{I}(T > t) - \hat{\pi}(t^* \mid \mathbf{Z}))^2 \right],$$

which measures the squared deviation between the observed survival status at time t and the predicted survival probability.

In the presence of right-censored observations, the Brier score cannot be computed directly and is therefore estimated using inverse probability of censoring weighting (IPC weighting) (Graf et al., 1999). This approach reweights observed contributions by the inverse of the estimated censoring distribution, ensuring unbiased estimation under independent censoring.

Null model and time horizon. To construct IPA, the Brier score of a prediction model is compared to the Brier score of a null model. The null model contains no covariates and represents baseline prediction. In the time-to-event setting, the null model survival function must be estimated for the same prediction horizon t .

In the absence of competing risks, the null model survival function is estimated using the Kaplan–Meier estimator. Let

$$t_{(1)} < t_{(2)} < \cdots < t_{(D)}$$

denote the ordered distinct observed event times. At each event time $t_{(k)}$, let d_k denote the number of observed events and n_k the number of individuals at risk immediately prior to $t_{(k)}$. The Kaplan–Meier estimator of the survival function according to Kaplan

and Meier (1958, p. 458-478), is given by

$$\hat{S}_{\text{KM}}(t) = \prod_{t_{(k)} \leq t} \left(1 - \frac{d_k}{n_k}\right).$$

Here, $n_k = |\{i : T_i \geq t_{(k)}\}|$ represents the size of the risk set at time $t_{(k)}$, and $d_k = |\{i : T_i = t_{(k)}, \Delta_i = 1\}|$ denotes the number of events occurring at that time. The product is taken over all distinct event times less than or equal to t . The estimator $\hat{S}_{\text{KM}}(t)$ therefore provides a nonparametric estimate of the marginal survival probability at time t under independent right-censoring.

Definition of IPA. Let $\text{BS}_{\text{model}}(t)$ denote the Brier score of a fitted prediction model at time t , and let $\text{BS}_{\text{null}}(t)$ denote the corresponding Brier score of the null model. The Index of Prediction Accuracy is defined as

$$\text{IPA}(t) = 1 - \frac{\text{BS}_{\text{model}}(t)}{\text{BS}_{\text{null}}(t)}.$$

By construction, IPA rescales the Brier score relative to the null model and yields a dimensionless measure of prediction accuracy.

Interpretation. IPA admits a simple and consistent interpretation. An IPA value of 1 corresponds to a perfect prediction model, while values less than or equal to 0 indicate a model that performs no better than, or worse than, the null model. Larger values of IPA indicate better predictive performance. Negative IPA values correspond to harmful models whose predictions are inferior to baseline prediction. IPA is used as the primary metric for comparing survival prediction models across different modeling strategies. By evaluating IPA at predefined prediction horizons and under a common null model, IPA enables a fair and interpretable comparison of Cox-based, tree-based, hybrid, and ensemble survival models.

3.7 Ensemble of Cox Models: Tree–Cluster Cox Model

Ensemble of Cox Models Method combines tree-based partitioning with Cox proportional hazards models in order to capture nonlinear covariate interactions while preserving interpretability. The approach follows the Tree–Cluster Cox framework proposed by

Shamsutdinova et al. (2022, p. 3-6), where a survival tree is used to partition the covariate space into clusters, and separate Cox models are fitted within each cluster.

Tree-based partitioning. In the first stage, a survival tree is grown using the training data. Unlike Random Survival Forests, the tree in this approach is not used directly for prediction. Instead, it serves as a clustering mechanism that partitions the covariate space into a finite number of disjoint regions (terminal nodes or leaves). The tree depth is restricted such that the total number of clusters does not exceed a predefined maximum.

Let $\mathcal{L} = \{1, \dots, L\}$ denote the set of terminal nodes of the tree. Each observation i with covariates $\mathbf{Z}_i$ is assigned to exactly one leaf $l(i) \in \mathcal{L}$ according to the splitting rules of the tree.

Cluster-specific Cox models. In the second stage, a separate Cox proportional hazards model is fitted within each leaf. For cluster $l \in \mathcal{L}$, the conditional hazard function is modeled as

$$\lambda_l(t \mid \mathbf{Z}) = \lambda_{0,l}(t) \exp(\boldsymbol{\beta}_l^\top \mathbf{Z}),$$

where $\lambda_{0,l}(t)$ is a cluster-specific baseline hazard function and $\boldsymbol{\beta}_l$ is a vector of regression coefficients estimated using only the observations assigned to cluster l .

This formulation relaxes the standard Cox model assumption of a common baseline hazard across all individuals by allowing different clusters to have distinct baseline hazard functions. In addition, fitting Cox models within homogeneous clusters simplifies the regression task and can improve stability when nonlinear effects or interactions are present in the full population.

Prediction. To obtain predictions for a new observation with covariates $\mathbf{Z}$, the observation is first passed through the survival tree to determine its cluster membership $l(\mathbf{Z})$. The corresponding cluster-specific Cox model is then used to compute the conditional hazard function

$$\lambda(t \mid \mathbf{Z}) = \lambda_{0,l(\mathbf{Z})}(t) \exp(\boldsymbol{\beta}_{l(\mathbf{Z})}^\top \mathbf{Z}),$$

where $\lambda_{0,l}(t)$ denotes the baseline hazard function for cluster l .

The corresponding predicted survival function is obtained as

$$S(t \mid \mathbf{Z}) = \exp\left(-\int_0^t \lambda_{0,l(\mathbf{Z})}(u) \exp(\boldsymbol{\beta}_{l(\mathbf{Z})}^\top \mathbf{Z}) du\right) = S_{0,l(\mathbf{Z})}(t)^{\exp(\boldsymbol{\beta}_{l(\mathbf{Z})}^\top \mathbf{Z})},$$

where $S_{0,l}(t)$ denotes the baseline survival function associated with $\lambda_{0,l}(t)$.

Equivalently, the overall survival function can be written as

$$S(t \mid \mathbf{Z}) = \sum_{l \in \mathcal{L}} \mathbb{I}\{\mathbf{Z} \in l\} S_{0,l}(t)^{\exp(\beta_l^\top \mathbf{Z})},$$

where $\mathbb{I}\{\mathbf{Z} \in l\}$ indicates membership of $\mathbf{Z}$ in cluster l .

Motivation and interpretability. The Tree-Cluster Cox approach addresses two limitations of the standard Cox model. First, it relaxes the assumption of a single baseline hazard function for the entire population by allowing cluster-specific baseline hazards. Second, it accommodates nonlinear effects and interactions through tree-based partitioning, while retaining the interpretability of Cox regression within each cluster.

3.8 Hybrid RSF-Cox Method

The hybrid method combines Random Survival Forests (RSF) with Cox proportional hazards models in order to exploit the strengths of both approaches. RSF is used to capture nonlinear effects and interactions and to assess variable importance, while the Cox model provides a structured and interpretable regression framework for survival prediction. The hybrid approach follows a two-stage procedure consisting of variable selection via RSF and subsequent survival modeling using a Cox model.

Variable ranking via RSF. In the first stage, a Random Survival Forest is fitted to the training data using the full set of available covariates. As described in the RSF subsection, the forest is grown using bootstrap sampling, random feature selection at each node, and survival-based splitting rules. After fitting the RSF, variable importance (VIMP) measures are computed based on out-of-bag (OOB) prediction error.

Let $\text{VIMP}(Z_k)$ denote the variable importance score for covariate Z_k . The variables are ranked according to their VIMP values, with larger values indicating greater predictive relevance. In this thesis, VIMP is used as a ranking heuristic rather than a definitive measure of causal importance, in line with the cautionary interpretation discussed in the RSF section.

Cox model fitting on selected variables. In the second stage, a subset of covariates is selected based on the RSF-derived VIMP ranking. Specifically, only the top N ranked variables are retained for subsequent modeling, where N is treated as a tuning parameter (or user-defined hyperparameter).

A Cox proportional hazards model is then fitted using this reduced set of covariates.

Let $\mathbf{Z}^{(\text{sel})}$ denote the vector of selected covariates. The hybrid Cox model is specified as

$$\lambda(t \mid \mathbf{Z}^{(\text{sel})}) = \lambda_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{Z}^{(\text{sel})}),$$

where $\lambda_0(t)$ is the baseline hazard function and $\boldsymbol{\beta}$ is the vector of regression coefficients estimated using partial likelihood. Depending on the modeling setting, the Cox model may be fitted either in its unpenalized form or with LASSO regularization to further stabilize estimation.

Prediction. For a new observation with covariates $\mathbf{Z}$, prediction proceeds by first restricting the covariate vector to the selected components $\mathbf{Z}^{(\text{sel})}$. The fitted Cox model is then used to compute the conditional hazard and survival functions in the standard Cox framework. In contrast to Tree-Cluster Cox Model, the RSF component in the hybrid method does not induce cluster-specific baseline hazards; instead, it serves exclusively as a data-driven variable selection mechanism.

Motivation and interpretation. The hybrid RSF-Cox method is motivated by the observation that RSF can effectively identify important variables and complex dependencies, while Cox models offer interpretability and stable estimation when applied to a suitably reduced feature set. By combining RSF-based variable screening with Cox regression, the hybrid approach aims to improve predictive performance relative to single-model approaches while retaining the interpretability of regression-based survival models.

3.9 k -fold Cross-Validation

k -fold cross-validation (Holling, 2009, p. 624) is used to assess the generalization performance of a model by partitioning the dataset into k equally sized folds. The model is trained and evaluated k times, such that each fold is used once as a test set while the

remaining $k-1$ folds form the training set. This procedure reduces bias arising from a single random data split.

Formally, a dataset D with n observations is randomly divided into k disjoint subsets $D_1, D_2, \dots, D_k$. In iteration i ($i = 1, \dots, k$), the model is trained on $D \setminus D_i$ and evaluated on D_i . Model performance is then obtained by aggregating the k test results according to the chosen evaluation metric.

The choice of k affects both estimation accuracy and computational cost. Smaller values of k tend to increase variance, whereas larger values (e.g. leave-one-out cross-validation with $k = n$) substantially increase computational burden. Common choices such as $k = 5$ or $k = 10$ are often used in practice, as they provide a balance between computational efficiency and estimation stability.

Nested (double) cross-validation. When model tuning or data-driven model selection is required, standard k -fold cross-validation can lead to optimistically biased performance estimates if the same data are used both to select hyperparameters and to evaluate predictive performance. To address this issue, *nested* (or *double*) cross-validation is employed. In nested cross-validation, an *outer* k -fold loop is used to estimate generalization performance, while an *inner* cross-validation loop is performed within each outer training fold to select tuning parameters or model components (e.g. the number of variables retained or tree depth). The model is then refitted on the full outer training set using the selected parameters and evaluated on the corresponding outer test fold. This separation ensures that the test data in the outer loop remain completely independent of all tuning decisions, yielding an approximately unbiased estimate of predictive performance. In this thesis, nested cross-validation is used throughout for all methods involving hyperparameter tuning or data-adaptive model construction.

3.10 Software Tools

All analyses were conducted using **R** (R Core Team, 2024), version 4.4.2. Survival modeling and inference were performed using the **survival** package (Therneau, 2026), which provides functions for fitting Cox proportional hazards models, computing Kaplan–Meier estimators, and handling right-censored survival data. Penalized regression models, including LASSO-regularized Cox regression, were fitted using **glmnet** (Friedman et. al, 2010), which supports efficient estimation in high-dimensional settings.

Random Survival Forests were implemented using `randomForestSRC` (Ishwaran and Kogalur, 2026), which was used for nonlinear survival modeling, variable importance estimation, and RSF-based variable ranking. Tree-based partitioning for the Tree-Cluster Cox method was performed with `rpart` (Therneau et al., 2013), allowing the construction of shallow survival trees for defining risk-based subgroups.

Model evaluation and performance assessment under right censoring were carried out using `riskRegression` (Gerds et. al, 2025) and `pec` (Gerds et. al, 2025). These packages were used to compute time-dependent Brier scores, Index of Prediction Accuracy (IPA), and related performance summaries within the nested cross-validation framework.

Data manipulation and preprocessing were performed using `tidyverse` (Wickham et al., 2023), especially `dplyr` and `tidyr`, which facilitated filtering, reshaping, and transformation of the analysis datasets. Results and intermediate outputs were exported using `openxlsx` (Schauberger and Walker, 2025) for further inspection and documentation.

4 Statistical Analyses

This section presents the statistical analyses, results with tables and figures, and their interpretation within the context of the study.

4.1 Descriptive Survival Results

A total of $n = 453$ patients were included in the descriptive survival analysis. During follow-up, 104 patients experienced the event of interest (death), corresponding to an event proportion of 22.8%, while 77.2% of observations were censored.

The overall Kaplan–Meier estimate of survival with death as the event of interest is shown in Figure 1.

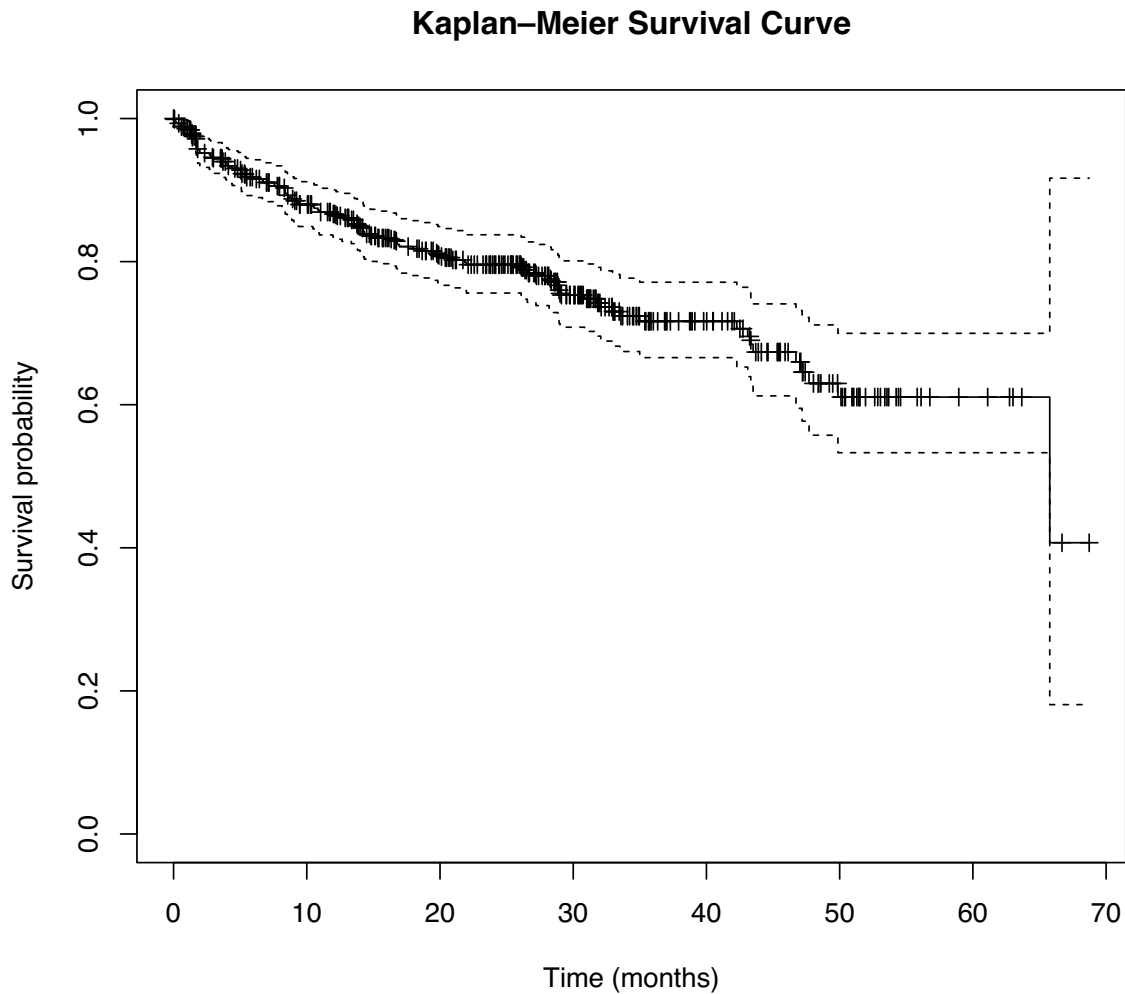


Figure 1: Overall Kaplan–Meier estimate of transplant-free survival with death as the event of interest, including pointwise 95% confidence intervals.

The estimated restricted mean survival time was 50.5 months (standard error 1.55 months). The median survival time was estimated at 65.8 months, however, the upper confidence limit could not be determined due to the high degree of censoring at longer follow-up times.

Observed survival times ranged from 0 to 68.7 months, with median and mean follow-up of 24.6 months and 23.7, respectively. This discrepancy occurs because the arithmetic mean of the observed follow-up times ignores right-censoring, while the restricted mean survival time accounts for censoring via the Kaplan–Meier estimator.

The distribution of survival times by event status is illustrated in Figure 2. When stratified by event status, patients who died during follow-up had substantially shorter observed survival times (median 10.7 months) compared with censored patients (median 26.7 months)

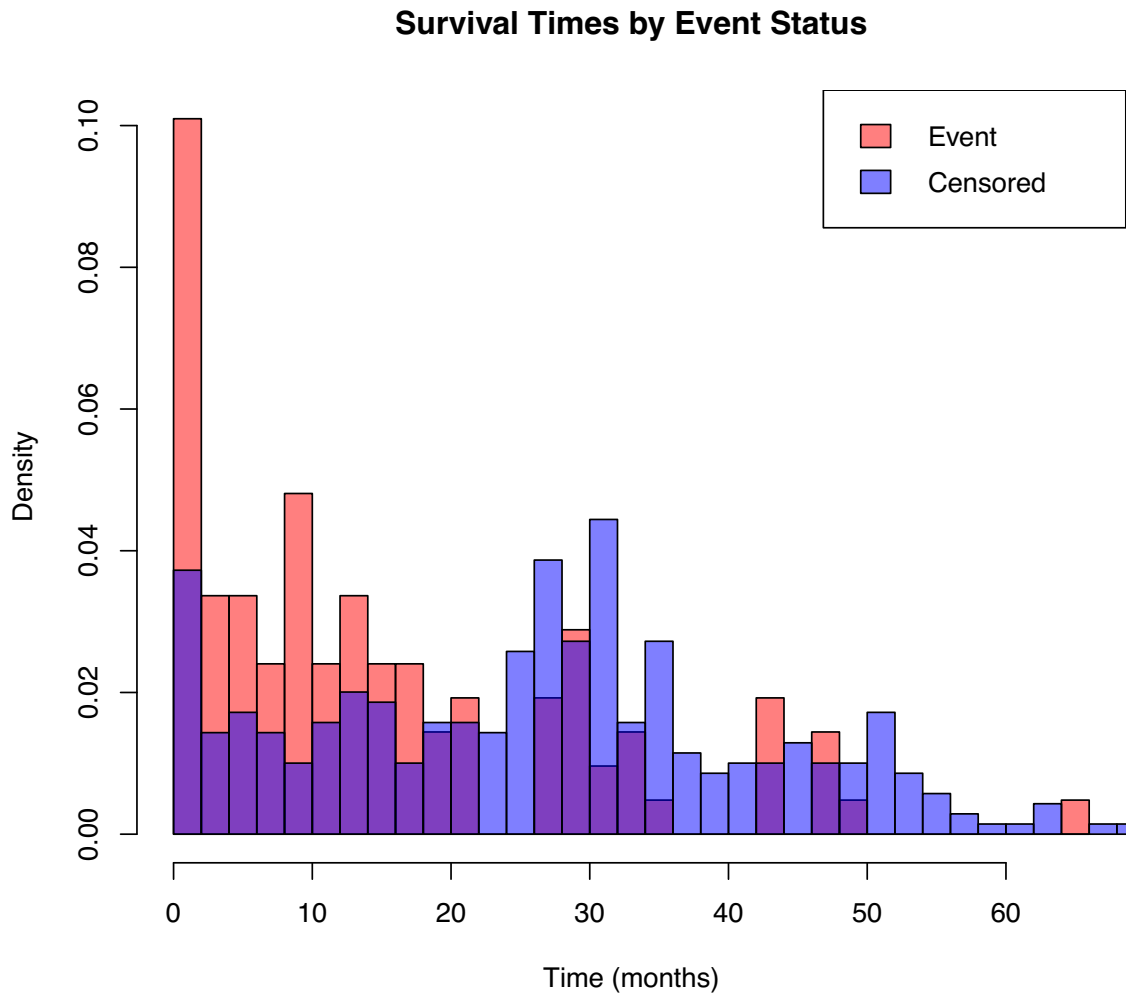


Figure 2: Histogramm of survival times by event.

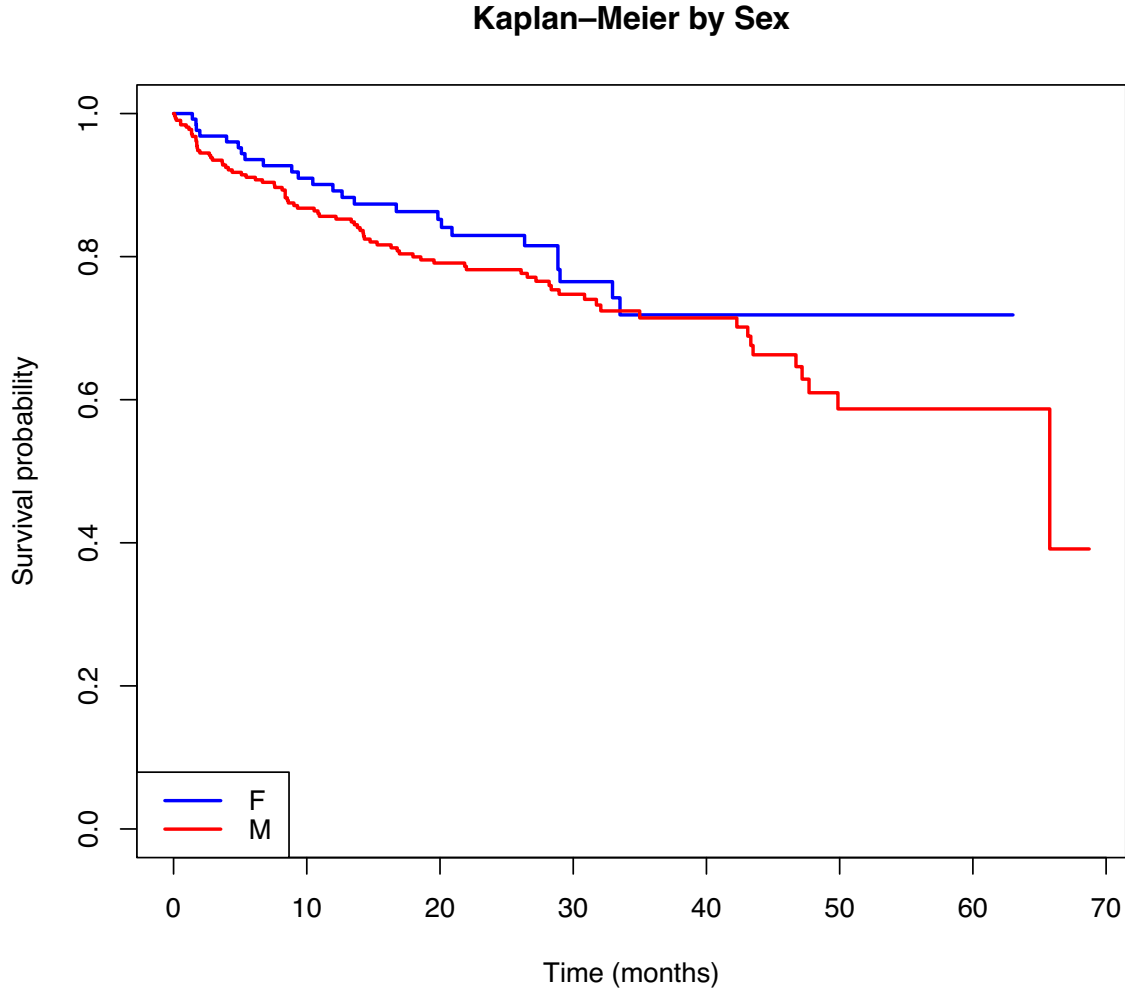


Figure 3: Kaplan-Meier estimate of transplant-free survival with death as the event of interest by sex (F= Female, M= Male).

Kaplan-Meier curves stratified by sex are shown in Figure 3. Female patients exhibited slightly higher survival probabilities over most of the follow-up period. However, a log-rank test comparing survival between sexes did not indicate a statistically significant difference. The log-rank test resulted in $\chi^2(1) = 1.10$, which is smaller than the critical value $\chi^2_{0.95,1} = 3.84$ at the 5% significance level. Therefore, the null hypothesis of equal survival distributions between male and female patients cannot be rejected ($p = 0.30$). Although minor differences in the Kaplan-Meier curves were observed between male and female patients, no statistically significant difference was detected based on the log-rank

test. These findings motivate the use of multivariable survival models to assess the joint predictive contribution of multiple predictors domain.

Handling of missing data. All subsequent analyses were conducted exclusively on a single imputed dataset. While Random Survival Forests are capable of handling missing values internally, Cox-based models require complete covariate information. To ensure comparability across modeling approaches, all analyses were therefore performed on the same imputed dataset. A complete-case analysis would have led to a considerable reduction in sample size and number of observed events, thereby reducing statistical power and potentially introducing selection bias..

Prior to imputation, variables with more than 14% missing values were excluded from the analysis, except for the survival outcome variables, which were retained regardless of missingness. The remaining variables were imputed using a deterministic imputation scheme applied uniformly across all predictor sets. Missing values in continuous variables were replaced by the median of the observed values, while missing values in categorical variables were replaced by the most frequent value (mode). Survival time and event indicators were not imputed.

The resulting completed dataset was used consistently for all subsequent modeling and evaluation steps to ensure comparability across methods.

4.2 LASSO-penalized Cox Model

This subsection reports the predictive performance of the LASSO-penalized Cox model evaluated using a nested cross-validation (double CV) framework. In the inner loop, the regularization parameter λ was tuned via $K_{\text{inner}} = 10$ -fold cross-validation using `cv.glmnet` (family = "cox", $\alpha = 1$). In the outer loop ($K_{\text{outer}} = 10$ folds), the fitted model was evaluated on held-out data using the Brier score (BS) at prediction horizons $t \in \{12, 24, 36, 48, 60\}$ months, restricted to the observed time range of the corresponding outer test fold. Predictive performance was summarized by the mean Brier score across horizons and by the Index of Prediction Accuracy (IPA), computed relative to the null Kaplan–Meier model as

$$\text{IPA} = 1 - \text{BS}/\text{BS}_{\text{null}}.$$

Here, n denotes the number of observations, d the number of observed death events, and p the number of predictors included in a given model.

LASSO-penalized Cox model formulation. For a given predictor combination, the survival outcome was modeled using a Cox proportional hazards model with an ℓ_1 -penalty (LASSO). Let T_i denote the transplant-free survival time for subject i , δ_i the event indicator ($\delta_i = 1$ for death, $\delta_i = 0$ for censoring), and $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})^\top$ the vector of p covariates.

The Cox model specifies the hazard function

$$h(t | \mathbf{x}_i) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}_i),$$

where $h_0(t)$ is an unspecified baseline hazard function and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$ are regression coefficients.

Estimation of $\boldsymbol{\beta}$ was performed by maximizing the penalized partial log-likelihood

$$\ell_{\text{pen}}(\boldsymbol{\beta}) = \ell_{\text{Cox}}(\boldsymbol{\beta}) - \lambda \sum_{j=1}^p |\beta_j|,$$

where $\ell_{\text{Cox}}(\boldsymbol{\beta})$ denotes the standard Cox partial log-likelihood and $\lambda \geq 0$ controls the degree of shrinkage. For $\lambda = 0$, the model reduces to the unpenalized Cox model, whereas increasing values of λ induce sparsity by shrinking coefficients toward zero.

Inner-loop selection of the regularization parameter. Within each outer cross-validation fold, the penalty parameter λ was tuned using an inner cross-validation procedure. Specifically, the value $\lambda_{\min}$ minimizing the mean cross-validated partial likelihood loss was selected and used to refit the model on the full outer training set. Consequently, both the selected variables and the effective model complexity were outer-fold specific. All fitted models and illustrative examples reported below correspond to Cox models estimated at $\lambda = \lambda_{\min}$.

Illustrative fitted model. As an illustrative example, consider one outer cross-validation fold of the `lab+ram` predictor combination, which includes routine laboratory measurements together with Raman spectroscopy variables. In this setting, the covariate vector can be written as for example,

$$\mathbf{x}_i = (\text{HB}_i, \text{BUN}_i, \text{CHE}_i, \text{CRP}_i, X_{202,i}, X_{203,i}, \dots, X_{3148,i})^\top,$$

where laboratory variables (e.g. hemoglobin, blood urea nitrogen, cholinesterase, C-reactive protein) are combined with Raman intensity features indexed by their spectral positions.

After penalization at $\lambda = \lambda_{\min}$, many coefficients are shrunk exactly to zero. The resulting fitted model in this outer fold takes the form

$$h(t \mid \mathbf{x}_i) = h_0(t) \exp(\beta_{\text{HB}} \text{HB}_i + \beta_{\text{BUN}} \text{BUN}_i + \beta_{\text{CRP}} \text{CRP}_i + \beta_{202} X_{202,i} + \beta_{3148} X_{3148,i}),$$

with all remaining coefficients equal to zero.

Predictive performance across predictor combinations. Table 4 summarizes results for all 15 predefined predictor-group combinations. Across all combinations, the null-model Brier score was stable ($\overline{\text{IBS}}_{\text{null}} \approx 0.181$), whereas model-based Brier scores and IPA varied substantially by predictor set.

Table 4: Nested cross-validation performance of the LASSO-penalized Cox model across predictor-group combinations. (p = nr. of predictors, n = nr. of obs., d = nr. of death events)

Combo ID	Groups	p	n	d	IPA_{mean}	IPA_{sd}	IBS_{mean}
1	lab	40	453	104	0.010	0.277	0.179
2	ba	17	453	104	-0.114	0.239	0.202
3	ram	1232	453	104	-0.058	0.231	0.191
4	clinical	22	453	104	0.025	0.253	0.176
5	lab+ba	57	453	104	0.017	0.271	0.178
6	lab+ram	1272	453	104	0.048	0.366	0.172
7	lab+clinical	62	453	104	0.045	0.295	0.172
8	ba+ram	1249	453	104	-0.021	0.189	0.184
9	ba+clinical	39	453	104	-0.033	0.288	0.187
10	ram+clinical	1254	453	104	-0.096	0.326	0.198
11	lab+ba+ram	1289	453	104	0.015	0.273	0.178
12	lab+ba+clinical	79	453	104	0.025	0.281	0.176
13	lab+ram+clinical	1294	453	104	-0.038	0.340	0.187
14	ba+ram+clinical	1271	453	104	-0.109	0.346	0.200
15	lab+ba+ram+clinical	1311	453	104	-0.094	0.364	0.197

The strongest mean performance was obtained for the *lab+ram* combination ($\text{IPA}_{\text{mean}} = 0.048$, $\text{IBS}_{\text{mean}} = 0.172$), followed by *lab+clinical* ($\text{IPA}_{\text{mean}} = 0.045$). However, performance variability across outer folds was pronounced, particularly for high-dimensional combinations, as reflected by large IPA standard deviations.

In contrast, bile-acid predictors alone and several combinations involving bile acids yielded negative mean IPA values, indicating worse average performance than the null Kaplan–Meier benchmark.

Selected penalty parameter. For the *lab+ram* combination, the selected $\lambda_{\min}$ values across outer folds had mean 0.0678 (SD 0.0188, median 0.0620, range [0.0518, 0.1123]). For the *lab+clinical* combination, the mean was 0.0680 (SD 0.0144, median 0.0663, range [0.0484, 0.0925]). These values reflect comparable average regularization strength, with slightly greater variability in the high-dimensional *lab+ram* setting.

Stability of variable selection. To assess the robustness of predictor selection, inclusion frequencies of variables selected by the LASSO ($\lambda = \lambda_{\min}$) were evaluated across the ten outer cross-validation folds. For each predictor combination, the number of folds in which a variable was retained was recorded. Only predictors selected in at least 50% of the outer folds (i.e., in at least 5 of 10 folds) are reported in Table 5.

For the *lab+clinical* combination, selection stability was high for the following variables: The laboratory predictors **bun**, **crp**, **che**, and **hb**, together with the clinical severity score **childscore**, were selected in at least 90% of outer folds. The indicator of hepatic encephalopathy severity (**bl_he_123severe**) was retained in 60% of folds. These findings indicate strong and consistent prognostic relevance of established laboratory and clinical markers.

For the *lab+ram* combination, the core laboratory predictors **hb**, **che**, **crp**, and **bun** again demonstrated high stability (90% inclusion). Among Raman spectroscopy features, **X958** was selected in 90% of folds, while **X2262** and **X3148** were retained in 70% of folds. Several additional Raman variables appeared in 50–60% of folds, indicating lower stability than the laboratory predictors.

Overall, laboratory and clinical variables exhibit strong selection stability across resampling iterations, while Raman-derived predictors contribute additional signal but show greater variability in inclusion frequency, likely reflecting the higher dimensionality and correlation of the spectral feature space.

Table 5: Predictors selected by the LASSO ($\lambda = \lambda_{\min}$) in at least 5 out of the $K_{\text{outer}} = 10$ outer folds.

Model	Predictor	Folds Selected	Proportion (%)
lab+clinical	bun	10	100
	childscore	10	100
	crp	10	100
	che	9	90
	hb	9	90
	bl_he_123severe	6	60
lab+ram	che	10	100
	crp	10	100
	hb	10	100
	X958	9	90
	bun	9	90
	X2262	7	70
	X3148	7	70
	X1843	6	60
	X1942	6	60
	X1944	6	60
	na	6	60
	X2208	5	50

4.3 Random Survival Forests

This subsection reports results for Random Survival Forests (RSF), a nonparametric ensemble method for right-censored survival data. In the outer loop ($K_{\text{outer}} = 10$ folds), predictive performance was assessed on held-out data using the Brier score (BS) at prediction horizons $t \in \{12, 24, 36, 48, 60\}$ months, restricted to the observed time range of each outer test fold. Performance was summarized by the mean Brier score across horizons and by the Index of Prediction Accuracy (IPA) relative to the null (Kaplan–Meier) model.

Nested cross-validation and hyperparameter tuning. RSF hyperparameters were tuned using an inner cross-validation loop ($K_{\text{inner}} = 5$ folds) nested within each outer training fold. The tuning grid comprised combinations of:

$$m_{\text{try}} \in \{2, 4, 8\}, \quad \text{nodesize} \in \{5, 10, 15\}, \quad \text{nsplit} = 10, \quad \text{ntree} = 500.$$

Within each inner fold, candidate RSF models were evaluated using the mean Brier score across the prespecified prediction horizons, and the hyperparameter combination minimizing this score was selected. The RSF was then refitted on the full outer training set using the selected hyperparameters and evaluated on the corresponding outer test fold.

To reduce computational burden, variable importance (VIMP) was computed only for the final outer-loop RSF fits using permutation importance, while importance calculations were disabled during inner-loop tuning.

Illustrative example. For the *lab+clinical* predictor combination, the fitted RSF model in a given outer fold can be viewed as an ensemble

$$\mathcal{F} = \{T_1, \dots, T_{500}\},$$

where each tree T_b is grown on a bootstrap sample of the outer training data using m_{try} randomly selected predictors at each split. Terminal nodes correspond to subgroups defined by recursive partitions of laboratory and clinical variables, and each node contains a Kaplan–Meier–type survival estimate.

For a new subject with covariate vector $\mathbf{x}_i$, the predicted survival probability at time t is given by

$$\hat{S}(t \mid \mathbf{x}_i) = \frac{1}{500} \sum_{b=1}^{500} \hat{S}_b(t \mid \mathbf{x}_i),$$

where $\hat{S}_b(t \mid \mathbf{x}_i)$ denotes the node-specific survival estimate from tree b .

Table 6 summarizes RSF performance across all 15 predefined predictor-group combinations. Across all models, the null-model Brier score was stable ($\overline{\text{IBS}}_{\text{null}} \approx 0.181$), while RSF models consistently achieved lower Brier scores and positive IPA values. The strongest mean performance was observed for the *lab+clinical* combination ($\text{IPA}_{\text{mean}} = 0.110$, $\text{IBS}_{\text{mean}} = 0.161$), followed closely by *lab+ba+ram+clinical* and *lab+ba+clinical*. In contrast to the LASSO-penalized Cox model, RSF achieved positive mean IPA values for all predictor combinations, indicating improvement over the Kaplan–Meier benchmark.

Table 6: Nested cross-validation performance of Random Survival Forests across predictor-group combinations. (p = nr. of predictors, n = nr. of obs., d = nr. of death events)

Combo ID	Groups	p	n	d	IPA _{mean}	IPA _{sd}	IBS _{mean}
1	lab	40	453	104	0.090	0.178	0.165
2	ba	17	453	104	0.023	0.146	0.177
3	ram	1232	453	104	0.061	0.141	0.170
4	clinical	22	453	104	0.073	0.166	0.168
5	lab+ba	57	453	104	0.092	0.168	0.164
6	lab+ram	1272	453	104	0.088	0.129	0.165
7	lab+clinical	62	453	104	0.110	0.158	0.161
8	ba+ram	1249	453	104	0.065	0.124	0.170
9	ba+clinical	39	453	104	0.058	0.232	0.171
10	ram+clinical	1254	453	104	0.077	0.110	0.167
11	lab+ba+ram	1289	453	104	0.075	0.127	0.168
12	lab+ba+clinical	79	453	104	0.093	0.181	0.164
13	lab+ram+clinical	1294	453	104	0.090	0.128	0.165
14	ba+ram+clinical	1271	453	104	0.075	0.115	0.168
15	lab+ba+ram+clinical	1311	453	104	0.095	0.132	0.164

Stability across outer folds. Despite overall positive mean performance, fold-wise IPA values exhibited substantial variability. For the *lab+clinical* combination, outer-fold IPA ranged from -0.17 to 0.36 , reflecting sensitivity to the particular training-test split. This variability is expected in tree-based ensemble methods applied to moderate sample sizes and highlights the importance of nested cross-validation for unbiased performance estimation.

Hyperparameter behavior. Across outer folds, the selected hyperparameters varied moderately. For the *lab+clinical* and *lab+ba+ram+clinical* combinations, the number of variables considered at each split (m_{try}) ranged between 2 and 8, while smaller terminal node sizes (`nodesize` = 5 or 10) were most frequently selected. This pattern suggests that relatively deep trees with aggressive splitting were favored to capture complex nonlinear interactions among predictors.

Stability of variable importance across outer folds. Because RSF VIMP is computed separately in each outer fold, stability across folds provides an empirical measure of how consistently the forest relies on specific predictors under resampling. To assess the robustness of predictor relevance in the RSF, permutation-based variable importance

(VIMP) was computed in each of the $K_{\text{outer}} = 10$ outer folds (using `importance="permute"` in the outer refit). For each fitted forest, predictors were ranked by VIMP and the top-10 variables were retained. Stability was quantified as the frequency with which a predictor appeared among the top-10 across outer folds. Predictors with presence in at least 70% of folds were considered *stable*. Table 7 summarizes predictors with stable variable importance across outer folds. The lab+clinical model shows a broader and more stable set of clinically established predictors, whereas the lab+ram model retains only a small subset of stable Raman features.

Table 7: Stability of top-10 permutation variable importance (VIMP) predictors across outer cross-validation folds for selected RSF models. A variable is reported if it appeared among the top-10 predictors in at least 70% of outer folds.

Model	Predictor	Folds in Top-10	Proportion (%)
lab+clinical	bun	10	100
	che	10	100
	childscore	10	100
	childstage	10	100
	hb	10	100
	il6	10	100
	bl_asc_123	7	70
	crp	7	70
lab+ram	il6	10	100
	che	9	90
	X960	8	80
	hb	7	70
lab+ram+clinical	il6	10	100
	childstage	9	90
	che	8	80
	childscore	8	80
	bl_asc_123	7	70
	hb	7	70
	X960	7	70

For the *lab+clinical* model, stability was high: **bun**, **che**, **hb**, **il6**, **childscore**, and **childstage** appeared in the top-10 in all 10 folds (100%), while **bl_asc_123** and **crp** appeared in 7/10 folds (70%). This indicates that the model repeatedly relies on a consistent set of laboratory and disease-severity markers when trained on different re-samples.

For the *lab+ram* model, the set of stable predictors was smaller: only **il6** (100%), **che** (90%), **X960** (80%), and **hb** (70%) met the stability threshold. The reduced number of

stable predictors suggests that adding high-dimensional Raman features increases competition among correlated predictors, resulting in a more variable ranking of importance across folds.

When Raman predictors were added on top of clinical information (*lab+ram+clinical*), the stable set remained dominated by clinical and laboratory variables: `il6` was stable in 100% of folds, `childstage` in 90%, `che` and `childscore` in 80%, and `bl_asc_123` and `hb` in 70%. Notably, only one Raman feature (`X960`) reached the stability threshold (70%). These findings suggest that Raman variables may contribute additional predictive signal, but only a limited subset exhibits consistent importance across resamples, whereas clinical severity indicators show substantially more robust relevance.

Overall, RSF provided more stable average predictive performance than the LASSO-penalized Cox model, particularly in high-dimensional settings, at the cost of reduced model interpretability. These findings motivate their inclusion as a strong nonlinear benchmark for subsequent ensemble and hybrid modeling approaches.

4.4 Tree–Cluster Cox Model

This subsection reports results for the *Tree–Cluster Cox Model* approach, a two-stage survival ensemble designed to combine (i) robust variable screening in high dimensions with (ii) locally adaptive, cluster-specific Cox modeling. In the outer loop ($K_{\text{outer}} = 10$ folds), predictive performance was evaluated on held-out data using the Brier score (BS) at prediction horizons $t \in \{12, 24, 36, 48, 60\}$ months, restricted to the observed time range of each outer test fold. Performance was summarized by the mean Brier score across horizons and by the Index of Prediction Accuracy (IPA) relative to the null (Kaplan–Meier) model.

Method implementation. Tree–Cluster Cox Method proceeds in two stages within each outer training fold:

1. **Stage 1: outer-fold RSF ranking (screening).** For each outer fold, a single Random Survival Forest (RSF) was fit on the *outer training* data using all candidate predictors in the current predictor-group combination. Permutation variable importance (`importance="permute"`) was computed and predictors were ranked in descending importance.

Importantly, this ranking was performed *separately within each outer fold*. Hence, no rankings were combined across folds and each outer fold produced its own ordered predictor list, which was then used exclusively within that fold for inner-loop tuning and final model fitting. This RSF was used *only* for ranking, it did not contribute directly to the final risk predictions.

2. Stage 2: survival tree clustering + Cox-per-cluster prediction model.

Given the ranked list from Stage 1 within a given outer fold, the top- n predictors were selected, where n is treated as a tuning hyperparameter (see tuning grid below in Paragraph 4.4).

These predictors were used to: (i) fit a shallow survival tree (via `rpart`, `method = "exp"`) that partitions the training set into a small number of clusters (terminal nodes), and (ii) fit a Cox model *within each cluster* using the same top- n predictors.

Clusters with fewer than `min_cluster_events=5` events were excluded from cluster-specific Cox fitting. To ensure predictions for all test subjects, a single *global* Cox model was also fit on the full outer training data using the same top- n predictors and used as a fallback if a subject fell into a cluster without a valid cluster-specific Cox model. If numerical issues still produced missing survival predictions, values were conservatively replaced by the Kaplan–Meier estimate from the stage-2 training data.

Nested cross-validation and hyperparameter tuning. Stage-2 hyperparameters were tuned using an inner cross-validation loop ($K_{\text{inner}} = 5$ folds) nested within each outer training fold. The tuning grid comprised combinations of:

$$\text{top_n} \in \{5, 10, 15\}, \quad \text{tree_maxdepth} \in \{1, 2\},$$

where `top_n` denotes the number of highest-ranked predictors retained from the outer-fold RSF ranking.

The optimal configuration was selected by minimizing the mean Brier score across the inner validation folds.

The following stage-2 parameters were kept fixed:

$$\text{minsplit} = 30, \quad \text{minbucket} = 15, \quad \text{cp} = 0.01, \quad \text{min_cluster_events} = 5.$$

Crucially, the RSF ranking was computed *once per outer fold* on the outer training set and reused throughout the inner-loop tuning. Thus, the inner loop tuned only the stage-2 model (tree depth and the number of top-ranked predictors), avoiding both computational overhead and potential information leakage from repeatedly refitting RSF models.

Illustrative example: model workflow. To illustrate the Tree-Cluster Cox Method, we consider an example based on a *bile-acid-dominant* predictor set selected by the outer-fold RSF ranking.

Stage 1: RSF-based variable screening. Using permutation-based variable importance from an RSF fit on the complete-case data, the top-10 predictors were identified as

$\{\text{ngal}, \text{cystatinc}, \text{tcdca}, \text{kim1}, \text{totalba}, \text{tca}, \text{totalba_muM}, \text{gcdca}, \text{gca}, \text{gudca}\}.$

This set combines established kidney injury markers with multiple bile-acid species, reflecting heterogeneous biochemical pathways highlighted by the RSF screening step.

Stage 2: Survival tree for risk stratification. Using these predictors, a shallow survival tree (`tree_maxdepth=2`) was fitted to partition patients into subgroups with distinct risk profiles (see example in Figure 4).

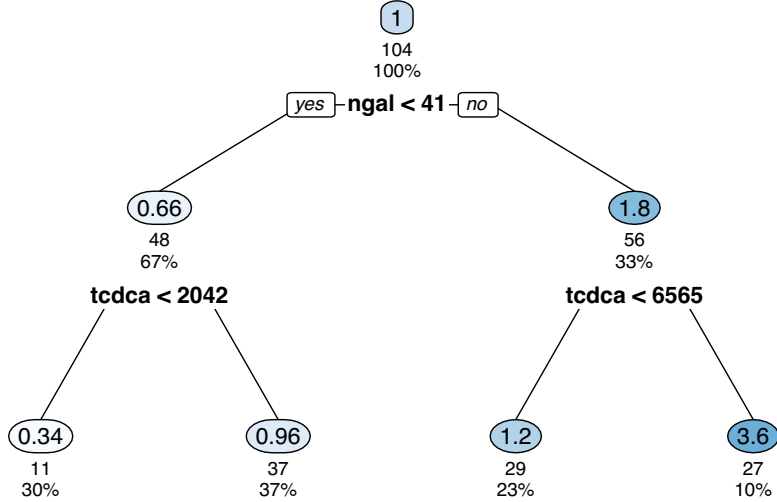


Figure 4: Stage-2 survival tree of the Tree-Cluster Cox Method ($\text{depth} \leq 2$).

The tree produced four terminal-node clusters, each corresponding to a region of the predictor space defined by combinations of renal injury and bile-acid levels. Cluster sizes ranged from $n = 46$ to $n = 169$, with event counts between 11 and 37, ensuring sufficient events for cluster-specific modeling.

Stage 3: Cluster-specific Cox models. Within each terminal node, a Cox proportional hazards model was fitted using the same top-10 predictors. No clusters were dropped due to insufficient events, and thus no fallback to a global Cox model was required in this illustrative fit.

Prediction for new observations. For a new subject with covariate vector $\mathbf{x}_i$, prediction proceeds as follows:

1. the subject is assigned to a terminal node via the survival tree,
2. the corresponding cluster-specific Cox model is selected, and
3. the survival probability at time t is computed as

$$\hat{S}(t \mid \mathbf{x}_i) = \hat{S}_{0,c(i)}(t)^{\exp(\mathbf{x}_i^\top \hat{\beta}_{c(i)})},$$

where $\hat{S}_{0,c(i)}(t)$ denotes the cluster-specific baseline survival.

Cluster-wise behavior of the Cox models (illustrative example). The cluster-specific Cox models show clear differences in which predictors drive risk across terminal-node clusters defined by the stage-2 survival tree (Figure 4). In Cluster 3 ($n = 136$, 11 events), corresponding to a subtree characterized by lower bile-acid levels, risk was mainly associated with the kidney injury marker `kim1`, while bile-acid predictors showed little evidence of association.

Cluster 4 ($n = 169$, 37 events), defined by higher values of renal dysfunction markers in the tree splits, was dominated by `cystatinc`, with other predictors contributing weakly or inconsistently. In Cluster 6 ($n = 102$, 29 events), arising from a branch combining renal and bile-acid-related splits, both marker groups contributed to risk: `cystatinc` remained the strongest kidney-related signal, while `tca` showed a moderate association. In contrast, Cluster 7 ($n = 46$, 27 events), which corresponds to a high bile-acid branch of the tree, showed no dominant renal signal, instead, several bile-acid species (`gudca`, `gca`, `gcdca`) were more prominent, though with greater uncertainty due to the smaller sample size. Full cluster-specific Cox model **outputs for this example are reported in Appendix B**. Overall, this example shows how the Tree-Cluster Cox Method uses

the survival tree to divide patients into groups with different clinical risks. It also then allows the dominant predictors within each clusters to differ, while maintaining a consistent Cox modeling framework within each cluster.

Results across predictor-group combinations. Table 8 summarizes Tree-Cluster Cox Method performance across all 15 predefined predictor-group combinations. The null-model Brier score was stable across combinations ($\overline{\text{IBS}}_{\text{null}} \approx 0.181$), while Tree-Cluster Cox Method achieved lower mean Brier scores for most combinations, corresponding to positive mean IPA values.

Table 8: Nested cross-validation performance of Tree-Cluster Cox Method across predictor-group combinations. (p = nr. of predictors, n = nr. of obs., d = nr. of death events)

Combo ID	Groups	p	n	d	IPA_{mean}	IPA_{sd}	IBS_{mean}
1	lab	40	453	104	0.096	0.213	0.163
2	ba	17	453	104	-0.070	0.216	0.195
3	ram	1232	453	104	0.072	0.116	0.168
4	clinical	22	453	104	0.021	0.247	0.177
5	lab+ba	57	453	104	0.092	0.206	0.164
6	lab+ram	1272	453	104	0.095	0.175	0.163
7	lab+clinical	62	453	104	0.046	0.210	0.172
8	ba+ram	1249	453	104	0.062	0.116	0.169
9	ba+clinical	39	453	104	0.007	0.258	0.180
10	ram+clinical	1254	453	104	0.051	0.256	0.171
11	lab+ba+ram	1289	453	104	0.066	0.291	0.169
12	lab+ba+clinical	79	453	104	0.014	0.300	0.178
13	lab+ram+clinical	1294	453	104	0.016	0.323	0.178
14	ba+ram+clinical	1271	453	104	0.064	0.206	0.169
15	lab+ba+ram+clinical	1311	453	104	0.043	0.308	0.173

The strongest mean performance was observed for *lab* ($\text{IPA}_{\text{mean}} = 0.096$; $\text{IBS}_{\text{mean}} = 0.163$) and *lab+ram* ($\text{IPA}_{\text{mean}} = 0.095$; $\text{IBS}_{\text{mean}} = 0.163$), followed closely by *lab+ba* ($\text{IPA}_{\text{mean}} = 0.092$; $\text{IBS}_{\text{mean}} = 0.164$). In contrast, the *ba*-only combination performed worse than the Kaplan-Meier benchmark ($\text{IPA}_{\text{mean}} = -0.070$).

Stability across outer folds and cluster formation. Fold-wise variability remained, reflecting moderate sample size and the additional instability introduced by learning both a partition (tree) and within-cluster Cox models. For example, for *lab+ram* (Combo 6), the outer-fold IPA ranged from negative values to strong improvements (e.g., $\text{IPA} \approx -0.23$ up to ≈ 0.37 in individual folds), highlighting sensitivity to the training–test split even under nested CV.

The stage-2 tree typically produced a small number of clusters (often 2 clusters for depth-1 trees, and up to 4 clusters when depth-2 trees were selected), with some folds yielding smaller clusters (e.g., $n \approx 16 - 57$ in the *lab+ram* model) however, all clusters retained sufficient event counts for cluster-specific Cox fitting, and no fallback to the global Cox model was required in this analysis.

Overall, Tree–Cluster Cox Method delivered consistent improvements over the null model when laboratory predictors were present, and it maintained positive mean performance in high-dimensional combinations involving Raman variables (e.g., *lab+ram*, *ram*, *ba+ram*). Conceptually, its gains arise from combining a high-dimensional screening step (RSF ranking) with a low-depth nonlinear partition that allows separate Cox effects across risk strata.

4.5 Hybrid RSF–Cox Method

This subsection reports results for the hybrid RSF–Cox approach, which combines Random Survival Forests (RSF) for data-driven variable screening with a global Cox proportional hazards model for survival prediction. As in the previous methods, predictive performance was assessed using a nested cross-validation design with $K_{\text{outer}} = 10$ outer folds. In each outer fold, performance was evaluated on held-out test data using the Brier score (BS) at prediction horizons $t \in \{12, 24, 36, 48, 60\}$ months, restricted to the observed time range of each outer test fold. Performance was summarized by the mean Brier score across horizons and by the Index of Prediction Accuracy (IPA) relative to the null (Kaplan–Meier) model.

Nested cross-validation and hyperparameter tuning. Within each outer training fold, a Random Survival Forest was fitted using all candidate predictors from the corresponding predictor-group combination. Permutation-based variable importance (VIMP)

was computed and used to rank predictors. This ranking was performed *once per outer fold* and held fixed throughout the corresponding inner cross-validation loop.

An inner cross-validation loop ($K_{\text{inner}} = 5$ folds) was used to tune the number of retained predictors,

$$\text{top_n} \in \{5, 10, 15, 20\}.$$

For each candidate value of `top_n`, a Cox proportional hazards model was fitted on the inner training folds using only the top-ranked predictors, and predictive performance was evaluated using the mean Brier score across the prespecified prediction horizons. The value of `top_n` minimizing the inner-loop mean Brier score was selected. Finally, a Cox model was refitted on the full outer training set using the selected predictors and evaluated on the corresponding outer test fold. This nested design ensures unbiased performance estimation and prevents information leakage from the test data into the variable selection step.

Illustrative behavior of variable selection. Across outer folds, the selected set of predictors varied moderately, reflecting sampling variability in the RSF-derived importance rankings. For example, in the *lab+ba+ram* combination (Combo 11), the optimal number of retained predictors ranged between `top_n` = 5 and 15 across folds. Despite this variability, a consistent core of clinically relevant laboratory variables (e.g., `il6`, `che`, `bun`, `hb`, `crp`) appeared frequently among the selected predictors, with Raman features entering the model in a fold-dependent manner. This behavior reflects the intended role of RSF in the hybrid approach: to provide a flexible, data-adaptive screening mechanism while retaining a single global regression model in the second stage.

Performance across predictor-group combinations. Table 9 summarizes hybrid model performance across all 15 predefined predictor-group combinations. Across models, the null-model Brier score was stable ($\overline{\text{IBS}}_{\text{null}} \approx 0.181$), enabling meaningful comparison of IPA values across combinations.

The strongest mean performance was observed for the *lab+ba+ram* combination ($\text{IPA}_{\text{mean}} = 0.112$; $\text{IBS}_{\text{mean}} = 0.160$), followed closely by the *lab* and *lab+ram* combinations. These results suggest that laboratory variables form a strong predictive backbone for the hybrid model, while the addition of selected bile-acid and Raman features can provide complementary prognostic information when filtered through RSF-based variable screening. Notably, the hybrid approach maintained stable performance even for very

Table 9: Nested cross-validation performance of the hybrid RSF–Cox method across predictor-group combinations. (p = nr. of predictors, n = nr. of obs., d = nr. of death events)

Combo ID	Groups	p	n	d	IPA _{mean}	IPA _{sd}	IBS _{mean}
1	lab	40	453	104	0.096	0.213	0.163
2	ba	17	453	104	−0.002	0.126	0.181
3	ram	1232	453	104	0.063	0.126	0.169
4	clinical	22	453	104	0.013	0.253	0.179
5	lab+ba	57	453	104	0.092	0.206	0.164
6	lab+ram	1272	453	104	0.095	0.175	0.163
7	lab+clinical	62	453	104	0.048	0.210	0.171
8	ba+ram	1249	453	104	0.062	0.116	0.169
9	ba+clinical	39	453	104	0.007	0.258	0.180
10	ram+clinical	1254	453	104	0.075	0.202	0.167
11	lab+ba+ram	1289	453	104	0.112	0.204	0.160
12	lab+ba+clinical	79	453	104	0.061	0.183	0.169
13	lab+ram+clinical	1294	453	104	−0.012	0.323	0.183
14	ba+ram+clinical	1271	453	104	0.064	0.206	0.169
15	lab+ba+ram+clinical	1311	453	104	0.022	0.296	0.177

high-dimensional feature sets (e.g., Raman spectra with $p > 1,200$), indicating that RSF-based variable ranking helps limit overfitting.

In contrast, predictor combinations relying primarily on bile-acid variables or on high-dimensional feature sets without laboratory measurements exhibited small or no improvement over the Kaplan–Meier benchmark. For these combinations, mean IPA values were close to zero and showed substantial fold-to-fold variability, suggesting that RSF-based screening alone may be insufficient to recover stable predictive structure in the absence of strong clinical or laboratory predictors.

Fold-wise variability in selected predictors. Inspection of the fold-specific selected predictors further illustrates the behavior of the hybrid RSF–Cox method. For the *lab+ba+ram* combination (Combo 11), the number of retained predictors selected in the inner loop varied between `top_n` = 5 and 15 across outer folds, reflecting uncertainty in the optimal model dimensionality under resampling. Despite this variability, a consistent core set of laboratory predictors was selected in nearly all folds, most notably `il6`, `che`, `bun`, `hb`, and `crp`.

Beyond this, the inclusion of Raman features (e.g., X950, X958, X960) varied substantially across folds, suggesting that their predictive contribution is sensitive to the particular training sample. This variability is mirrored in fold-wise performance: while several outer folds achieved substantial improvements over the null model (IPA up to 0.35), others yielded near-zero or negative IPA values. Taken together, these results indicate that the hybrid approach can leverage additional high-dimensional information when it is informative, but that the associated gains are not uniformly stable across resamples.

Overall, hybrid model performance changed gradually across predictor-group combinations, without sharp improvements or breakdowns. This behavior reflects the design of the hybrid approach, where RSF is used to screen for informative variables, while the final survival model remains a single global Cox model, yielding stable and interpretable predictions.

4.6 Comparison of Modeling Approaches

This subsection provides a comparative assessment of the four survival modeling approaches considered in this study, LASSO-penalized Cox regression, Random Survival Forests (RSF), Tree-Cluster Cox, and the Hybrid RSF-Cox approach. Comparison is based on the mean Index of Prediction Accuracy (IPA) obtained from nested cross-validation across all predefined predictor-group combinations.

Overall comparison across predictor groups. Table 10 summarizes the mean IPA values achieved by each modeling approach across all 15 predictor-group combinations. For each combination, the best-performing method and its corresponding mean IPA value are reported.

Across nearly all predictor groups, tree-based or hybrid methods outperformed the LASSO-penalized Cox model, which frequently yielded near-zero or negative IPA values, particularly for high-dimensional feature sets. This indicates limited ability of a single linear model with global regularization to capture the complex structure present in the data.

Random Survival Forests (RSF) achieved the highest mean IPA in 9 out of the 15 predictor combinations. RSF was the best-performing method for combinations involving clinical predictors, including C (IPA_{mean} = 0.073), $L+C$ (0.110), $B+C$ (0.059), $R+C$ (0.077), and $L+B+C$ (0.093), as well as for the full combination $L+B+R+C$ (0.095).

Table 10: Comparison of mean IPA values across modeling approaches and predictor-group combinations. Predictor groups are abbreviated as follows: L = laboratory, B = bile acids, R = Raman, C = clinical. For each combination, the best-performing method is reported.

Combo	Groups	LASSO	RSF	Tree-Cox	Hybrid	Best IPA	Best Method
1	L	0.010	0.090	0.096	0.096	0.096	Hybrid
2	B	-0.114	0.023	-0.070	-0.002	0.023	RSF
3	R	-0.058	0.061	0.073	0.063	0.073	Tree-Cox
4	C	0.025	0.073	0.021	0.013	0.073	RSF
5	L+B	0.017	0.092	0.092	0.092	0.092	Hybrid
6	L+R	0.048	0.088	0.095	0.095	0.095	Tree-Cox
7	L+C	0.045	0.110	0.046	0.048	0.110	RSF
8	B+R	-0.021	0.065	0.062	0.062	0.065	RSF
9	B+C	-0.033	0.059	0.007	0.007	0.059	RSF
10	R+C	-0.096	0.077	0.051	0.075	0.077	RSF
11	L+B+R	0.015	0.075	0.066	0.112	0.112	Hybrid
12	L+B+C	0.025	0.093	0.014	0.061	0.093	RSF
13	L+R+C	-0.038	0.089	0.016	-0.012	0.089	RSF
14	B+R+C	-0.109	0.075	0.064	0.064	0.075	RSF
15	L+B+R+C	-0.094	0.095	0.043	0.022	0.095	RSF

These results highlight the strong performance of fully nonlinear survival forests when clinical information is present.

Tree-Cluster Cox achieved the highest mean IPA for combinations dominated by Raman features, specifically R ($\text{IPA}_{\text{mean}} = 0.073$) and $L+R$ (0.095). This suggests that localized risk stratification via tree-based partitioning can be beneficial in settings where high-dimensional biochemical predictors contribute heterogeneous signal.

The Hybrid RSF-Cox approach performed best for the L ($\text{IPA}_{\text{mean}} = 0.096$), $L+B$ (0.092), and $L+B+R$ (0.112) combinations. Notably, the hybrid method achieved the single highest mean IPA observed across all models and predictor groups for $L+B+R$, indicating that RSF-based variable screening followed by a global Cox model can effectively leverage complementary information from the laboratory and Raman predictors.

Pairwise comparison of predictive performance. To further examine differences between modeling approaches, pairwise scatter plots of mean IPA values were constructed for all method combinations. Figure 5 summarizes these comparisons in a single multi-panel display.

In each panel, each point corresponds to one predictor-group combination, and the diagonal line indicates equal predictive performance between the two methods.

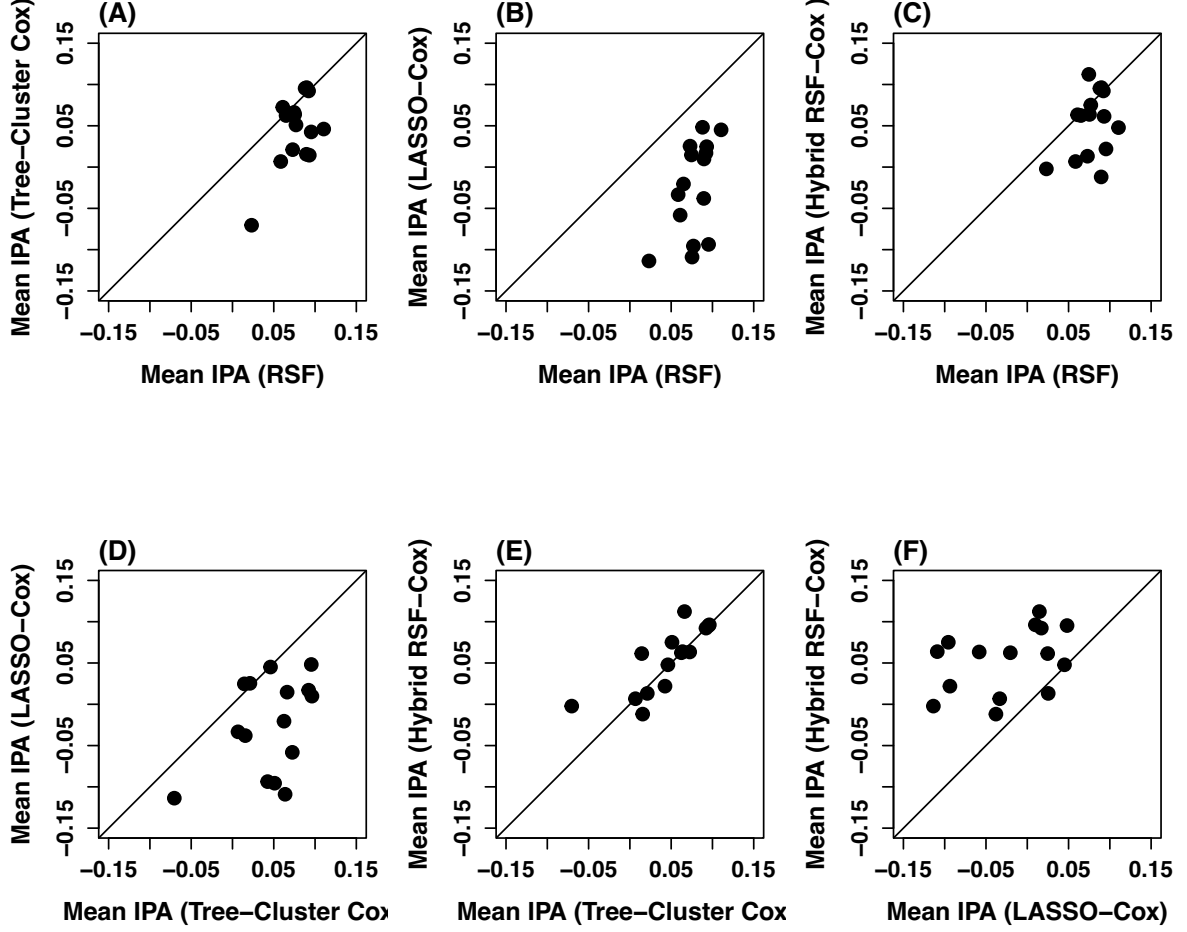


Figure 5: Pairwise comparisons scatter plots of mean IPA values for all methods.

Panels (A)–(C) compare Random Survival Forests (RSF) against Tree-Cluster Cox, the LASSO-penalized Cox model, and the Hybrid RSF-Cox method, respectively. In these comparisons, most points lie below the diagonal, indicating that RSF achieves higher IPA values for the majority of predictor-group combinations.

Panels (D) & (F) compare Tree-Cluster Cox against the LASSO-penalized Cox model and the Hybrid approach, respectively, and Panel (E) compares the Hybrid approach against the LASSO-penalized Cox model. In comparison between the LASSO-penalized Cox model and the Tree-Cluster Cox model, the latter approach outperforms the LASSO-

penalized model in almost all group combinations, while some points lie on the diagonal. On the other hand, in Panel (E), the Hybrid method’s performance is comparable to that of the Tree-Cox Model. Lastly, in Panel (F), the hybrid approach outperforms the LASSO-penalized model in almost all group combinations. This strongly suggests that the LASSO-penalized Cox model is the weakest among all 4 approaches. Overall, RSF is the best method among the 4 methods, especially when dealing with combinations that include the clinical predictors domain. The hybrid method is the second-best approach, especially when dealing with a high-dimensional predictor domain like *raman*.

5 Summary and Discussion

This thesis explored and compared several statistical methods for predicting survival in patients with liver disease. The analysis used a clinical dataset of 457 patients, which included right-censored survival times and a wide range of predictors. Available covariates covered several domains: laboratory measurements, clinical variables, bile-acid profiles, and high-dimensional Raman spectroscopic features.

To assess the contribution of these data sources, 15 predefined groups of predictors were created and evaluated. Four modeling strategies were considered: a Cox proportional hazards model with LASSO regularization, Random Survival Forests (RSF), a hybrid RSF–Cox approach, and a tree-based ensemble method that combines survival trees with cluster-specific Cox models (Tree-Cluster Cox). Model performance was assessed using nested cross-validation, with an inner loop for tuning of hyperparameters and an outer loop for performance estimation. Predictive accuracy was measured using the time-dependent Brier score and the Integrated Prediction Accuracy (IPA), with all results reported as cross-validated means.

In almost all predictor combinations, tree-based and hybrid methods performed better than the LASSO-penalized Cox model. The LASSO method frequently produced near-zero or negative IPA values, especially for high-dimensional predictor sets like Raman features. This shows that a global linear model with regularization has limited ability to approximate complex prognostic structures. For example, when Raman predictors were used alone, the LASSO model had a mean IPA of -0.058 , compared to 0.061 for RSF and 0.073 for Tree-Cluster Cox. Laboratory predictors were the most consistently informative type. When used alone, laboratory variables gave stable and positive IPA values across tree-based and hybrid approaches, with the hybrid RSF–Cox method

yielding a mean IPA of 0.096. When laboratory predictors were combined with other data sources, predictive performance generally improved. In particular, the combination of laboratory, bile-acid, and Raman predictors (*lab+ba+raman*) achieved the highest mean IPA across all models and predictor sets ($\text{IPA}_{\text{mean}} = 0.112$), obtained using the hybrid approach.

Clinical predictors also made a strong contribution to predictive performance, especially with nonlinear modeling. Random Survival Forests achieved a mean IPA of 0.073 for clinical variables alone and 0.110 for the laboratory and clinical combination, representing some of the strongest results observed for RSF. This pattern suggests that clinical variables involve nonlinear effects or interactions that linear regression-based survival models do not capture well. In contrast, bile-acid predictors had limited value on their own. For bile-acid variables alone, mean IPA values stayed modest across all methods, with RSF achieving an IPA of 0.023 and other methods producing values close to zero or negative.

However, bile-acid predictors added value when combined with laboratory or clinical variables. This suggests their prognostic value depends on other variables rather than being independent. Notably, Raman-derived predictors did not consistently improve results when used alone, but they did contribute to predictive performance when combined with laboratory or clinical variables, mainly within tree-based or hybrid modeling frameworks. While Raman-only combinations achieved moderate performance with the Tree-Cluster Cox method ($\text{IPA}_{\text{mean}} = 0.073$), their biggest contribution was seen in multimodal settings such as *lab+ba+raman*, where they were part of the best-performing predictor combination overall. This suggests that Raman features capture complementary information rather than serving as strong standalone prognostic markers.

When comparing the four modeling methods directly, Random Survival Forests stood out as the strongest overall method in this study. RSF had the highest mean IPA in 9 out of 15 predictor-group combinations and showed consistently positive performance across low- and high-dimensional settings. The hybrid RSF–Cox approach ranked second overall. While it was the best-performing method in fewer combinations, it achieved the single highest mean IPA in the study ($\text{IPA}_{\text{mean}} = 0.112$ for *lab+ba+raman*).

The Tree-Cluster Cox method performed best in Raman-heavy settings, such as *raman* and *lab+raman*, but showed more variability across resampling folds, reflecting sensitivity to the learned tree structure. The LASSO-penalized Cox model showed the weakest overall performance, rarely outperforming tree-based or hybrid methods and consistently

underperformed under high-dimensional predictor spaces. Pairwise comparisons based on IPA showed that the relative ranking of methods remained mostly consistent across performance measures.

Several limitations should be noted. Many model comparisons were performed without formal adjustment for multiple testing, so the results should be interpreted as descriptive and comparative rather than inferential. In addition, the analysis was based on a single dataset, limiting the generalization of the findings.

Some predictor combinations, especially those featuring high-dimensional features, showed substantial variability between folds, indicating sensitivity to resampling and limited effective sample size. Despite these limitations, the results show that flexible survival modeling methods can substantially improve prognostic performance in complex clinical data. Random Survival Forests provide strong and robust overall performance, while hybrid and ensemble strategies grant complementary strengths by keeping flexibility, interpretability, and stability. Future work ought to focus on external validation in independent cohorts, developing alternative ensemble strategies, and further investigation of the clinical usefulness of the resulting risk predictions.

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Appendix

Appendix A: Variable Dictionary

This appendix provides a comprehensive overview of all variables included in the dataset. For each variable, the table reports the abbreviated variable name used in the analysis, its full description, and the corresponding unit of measurement where applicable. Variables are grouped according to their thematic category.

A.1 Clinical Characteristics

Table 11: Clinical characteristics.		
Variable	Description	Unit
SEX	Sex (m = male, f = female)	–
SEX_Code	Sex (0 = female, 1 = male)	–
AGE	Age	years
HVPG	Hepatic venous pressure gradient	mmHg
HVPG_3gr_16	HVPG groups (6–9, 10–15, ≥ 16)	–
HVPG_3gr_20	HVPG groups (6–9, 10–19, ≥ 20)	–
HVPG_2gr_16	HVPG groups (6–15, ≥ 16)	–
EASL_BL_5	EASL stage at baseline (0–5)	–
BL_dACLD_01	Decompensated ACLD at baseline	–
VARBLEED_01	Previous variceal bleeding	–
ASC_01	Previous ascites	–
HE_01	Previous hepatic encephalopathy	–
HCC_BL_01	Hepatocellular carcinoma at baseline	–
ETIO_systematic	Etiology subclassification	–
ETIO_Code	Numeric etiology code	–
BL_STIFFNESS	Liver stiffness (FibroScan)	kPa
VARICES_012	Varices (0=no, 1=yes, 2=large)	–
BL_ASC_123	Baseline ascites severity (1–3)	–
BL_HE_123	Baseline HE severity (1–3)	–
CHILD_score	Child–Pugh score	unitless
CHILD_stage	Child–Pugh stage (A–C)	–
MELD	MELD score	unitless

Variable	Description	Unit
CNI_01	Chronic kidney disease	–
BL_DM_012	Baseline diabetes mellitus	–
BL_BMI	Body mass index	kg/m ²
ICG-PDR	Indocyanine green plasma disappearance rate	%/min
PCG-R15	Plasma clearance rate at 15 minutes	%

A.2 Laboratory Measurements

Table 12: Laboratory measurements.

Variable	Description	Unit
HB	Hemoglobin	g/dL
PLT	Platelets	10 ⁹ /L
WBC	White blood cell count	10 ⁹ /L
NA	Sodium	mmol/L
K	Potassium	mmol/L
CL	Chloride	mmol/L
CREA	Creatinine	mg/dL
BUN	Blood urea nitrogen	mg/dL
BILI	Bilirubin	mg/dL
PROTEIN	Total protein	g/L
ALB	Albumin	g/L
CHE	Cholinesterase	U/L
AP	Alkaline phosphatase	U/L
ASAT	Aspartate aminotransferase (AST)	U/L
ALAT	Alanine aminotransferase (ALT)	U/L
GGT	Gamma-glutamyl transferase	U/L
LDH	Lactate dehydrogenase	U/L
AAMY	Alpha-amylase	U/L
PAMY	Pancreatic amylase	U/L
LIP	Lipase	U/L
VWF	Von Willebrand factor	%
FGEN	Fibrinogen	mg/dL
INR	International normalized ratio	ratio

Variable	Description	Unit
BA_MUW	Bile acids (MUW laboratory)	$\mu\text{mol/L}$
TRIG	Triglycerides	mg/dL
CHOL	Cholesterol	mg/dL
GLUC	Glucose	mg/dL
HDL	High-density lipoprotein cholesterol	mg/dL
LDL	Low-density lipoprotein cholesterol	mg/dL
HbA1c	Hemoglobin A1c	%
NH3	Ammonia	$\mu\text{mol/L}$
CRP	C-reactive protein	mg/dL
IL6	Interleukin-6	pg/mL
PCT	Procalcitonin	ng/mL
LBP	Lipopolysaccharide-binding protein	$\mu\text{g/mL}$
IgA	Immunoglobulin A	mg/dL
IgG	Immunoglobulin G	mg/dL
IgM	Immunoglobulin M	mg/dL
IgG_1	Immunoglobulin G subclass 1	mg/dL
IgG_2	Immunoglobulin G subclass 2	mg/dL
IgG_3	Immunoglobulin G subclass 3	mg/dL
IgG_4	Immunoglobulin G subclass 4	mg/dL
COPEPTIN	Copeptin	pmol/L
BNP	Brain natriuretic peptide	pg/mL
ELF	Enhanced liver fibrosis score	unitless
TIMP1	Tissue inhibitor of metalloproteinase 1	ng/mL
P3NP	Procollagen type III N-terminal peptide	ng/mL
HA	Hyaluronic acid	ng/mL

A.3 Bile Acid Profile (Absolute Concentrations)

Table 13: Bile acid profile variables (absolute concentrations) and related biomarkers.

Variable	Description	Unit
BAPROF	Bile acid profile indicator	unitless
Sample	Biological sample identifier	–
Biobank_Nr	Biobank sample number	–

Variable	Description	Unit
Biobank_Kerstin	Biobank reference (Kerstin)	–
TMCA	Tauromuricholic acid	nmol/L
TCA_sulfate	Taurocholic acid sulfate	nmol/L
TUDCA	Tauroursodeoxycholic acid	nmol/L
TCA	Taurocholic acid	nmol/L
oMCA	Omega muricholic acid	nmol/L
GCA	Glycocholic acid	nmol/L
GUDCA	Glycoursodeoxycholic acid	nmol/L
CA_sulfate	Cholic acid sulfate	nmol/L
aMCA	Alpha muricholic acid	nmol/L
bMCA	Beta muricholic acid	nmol/L
TCDCA	Taurochenodeoxycholic acid	nmol/L
TDCA	Taurodeoxycholic acid	nmol/L
CA	Cholic acid	nmol/L
GCDCA	Glycochenodeoxycholic acid	nmol/L
UDCA	Ursodeoxycholic acid	nmol/L
GDCA	Glycodeoxycholic acid	nmol/L
TLCA	Taurolithocholic acid	nmol/L
CDCA	Chenodeoxycholic acid	nmol/L
GLCA	Glycolithocholic acid	nmol/L
DCA	Deoxycholic acid	nmol/L
LCA	Lithocholic acid	nmol/L
C4	7 α -Hydroxy-4-Cholesten-3-One (C4)	nmol/L
BA_total_nmol_L	Total bile acids	nmol/L
BA_total_μM	Total bile acids	μmol/L
KIM1	Kidney injury molecule-1	pg/mL
NGAL	Neutrophil gelatinase-associated lipocalin	ng/mL
CystatinC	Cystatin C	ng/mL

A.4 Raman Spectroscopy Variables

The Raman spectroscopy variables consist of high-dimensional continuous predictors indexed from `x202` to `x3148`. Each variable represents the spectral intensity measured at a specific Raman shift. All Raman variables are fully observed and were used without imputation.

A.5 Follow-up Events and Survival Endpoints

Table 14: Follow-up event indicators and time-to-event variables.

Variable	Description	Unit
FU_EVENTS	Number of follow-up events	count
HEA_Months	Time to first overt HE	months
HEA_01	Overt HE (0 = No, 1 = Yes)	–
RecHE_Months	Time to recurrent HE	months
RecHE_01	Recurrent HE (0 = No, 1 = Yes)	–
AKI_Months	Time to acute kidney injury	months
AKI_01	Acute kidney injury (0 = No, 1 = Yes)	–
HRS AKI_Months	Time to HRS–AKI	months
HRS AKI_01	HRS–AKI (0 = No, 1 = Yes)	–
INF_Months	Time to first infection	months
INF_01	Infection (0 = No, 1 = Yes)	–
FirDEC_cACLD_Months	Time to first cACLD decompensation	months
FirDEC_cACLD_01	First cACLD decompensation (0/1)	–
FurDEC_cACLD_Months	Time to further cACLD decompensation	months
FurDEC_cACLD_01	Further cACLD decompensation (0/1)	–
FurDEC_dACLD_Months	Time to further dACLD decompensation	months
FurDEC_dACLD_01	Further dACLD decompensation (0/1)	–
ACLF_Months	Time to ACLF	months
ACLF_01	ACLF (0 = No, 1 = Yes)	–
LRD_Months	Time to liver-related death	months
LRD_013	(0 = No, 1=LR, 3= non-LR)	–
ACLF_LRD_Months	Time to ACLF or LR death	months
ACLF_LRD_01	ACLF or LR death (0/1)	–
CUMULATIVE_Months	Time to composite progression	months

Variable	Description	Unit
CUMULATIVE_01	Composite progression (0/1)	–
REG_dACLD_strict_Months	Time to strict dACLD regression	months
REG_dACLD_strict_01	Strict dACLD regression (0/1)	–
TFS_months	Transplant-free survival time	months
TFS_Event_012	(0 = cens., 1 = Tx, 2 = death)	–

Appendix B: Section 4.4 Cluster-Specific Example Cox Model Outputs

This appendix reports the full outputs of the cluster-specific Cox proportional hazards models fitted in the illustrative example of the Tree-Cluster Cox Method (Section 4.4). These results are provided for completeness and transparency. As emphasized in the main text, the illustrative example is intended to demonstrate model mechanics rather than to support inferential conclusions; coefficient estimates and test statistics should therefore be interpreted with caution, particularly in clusters with limited numbers of events.

Cluster 3 ($n = 136$, **events** = 11)

```
=====
Cluster: 3
=====

Call:
survival::coxph(formula = f_cox, data = dcl, x = TRUE)

n= 136, number of events= 11

              coef exp(coef) se(coef)      z Pr(>|z|)
ngal         -2.714e-02  9.732e-01  3.289e-02 -0.825  0.40924
cystatinc    -1.400e-03  9.986e-01  1.532e-03 -0.914  0.36094
tcdca        -4.478e-04  9.996e-01  1.037e-03 -0.432  0.66590
kim1          8.008e-04  1.001e+00  3.089e-04  2.592  0.00953 **
totalba       7.010e-05  1.000e+00  1.872e-04  0.375  0.70803
tca           1.554e-04  1.000e+00  1.263e-03  0.123  0.90206
totalba_muM      NA         NA  0.000e+00    NA      NA
gcdca        -1.123e-04  9.999e-01  3.530e-04 -0.318  0.75039
gca          -9.352e-05  9.999e-01  5.655e-04 -0.165  0.86864
gudca        -2.162e-04  9.998e-01  5.002e-04 -0.432  0.66559
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
ngal	0.9732	1.0275	0.9125	1.038
cystatinc	0.9986	1.0014	0.9956	1.002
tcdca	0.9996	1.0004	0.9975	1.002
kim1	1.0008	0.9992	1.0002	1.001
totalba	1.0001	0.9999	0.9997	1.000
tca	1.0002	0.9998	0.9977	1.003
totalba_muM	NA	NA	NA	NA
gcdca	0.9999	1.0001	0.9992	1.001
gca	0.9999	1.0001	0.9988	1.001
gudca	0.9998	1.0002	0.9988	1.001

Concordance= 0.644 (se = 0.095)

Likelihood ratio test= 7.27 on 9 df, p=0.6

Wald test = 10.76 on 9 df, p=0.3

Score (logrank) test = 20.38 on 9 df, p=0.02

Cluster 4 (*n* = 169, events = 37)

=====

Cluster: 4

=====

Call:

survival::coxph(formula = f_cox, data = dcl, x = TRUE)

n= 169, number of events= 37

	coef	exp(coef)	se(coef)	z	Pr(> z)
ngal	-2.866e-02	9.717e-01	1.791e-02	-1.601	0.109444
cystatinc	1.863e-03	1.002e+00	5.315e-04	3.505	0.000457 ***
tcdca	-1.261e-05	1.000e+00	2.782e-05	-0.453	0.650209
kim1	-3.567e-05	1.000e+00	5.470e-04	-0.065	0.948012
totalba	3.886e-06	1.000e+00	1.445e-05	0.269	0.787999
tca	1.110e-05	1.000e+00	1.090e-04	0.102	0.918879
totalba_muM	NA	NA	0.000e+00	NA	NA

gcdca	9.993e-06	1.000e+00	2.308e-05	0.433	0.665036
gca	-1.935e-05	1.000e+00	5.308e-05	-0.365	0.715459
gudca	-8.509e-06	1.000e+00	1.849e-05	-0.460	0.645398

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ngal	0.9717	1.0291	0.9382	1.006
cystatinc	1.0019	0.9981	1.0008	1.003
tcdca	1.0000	1.0000	0.9999	1.000
kim1	1.0000	1.0000	0.9989	1.001
totalba	1.0000	1.0000	1.0000	1.000
tca	1.0000	1.0000	0.9998	1.000
totalba_muM	NA	NA	NA	NA
gcdca	1.0000	1.0000	1.0000	1.000
gca	1.0000	1.0000	0.9999	1.000
gudca	1.0000	1.0000	1.0000	1.000

Concordance= 0.665 (se = 0.051)

Likelihood ratio test= 13.74 on 9 df, p=0.1

Wald test = 15.11 on 9 df, p=0.09

Score (logrank) test = 15.89 on 9 df, p=0.07

Cluster 6 ($n = 102$, events = 29)

=====

Cluster: 6

=====

Call:

survival::coxph(formula = f_cox, data = dcl, x = TRUE)

n= 102, number of events= 29

	coef	exp(coef)	se(coef)	z	Pr(> z)
ngal	-4.526e-03	9.955e-01	3.101e-03	-1.459	0.1445

cystatinc	4.764e-04	1.000e+00	2.201e-04	2.165	0.0304 *
tcdca	-3.080e-04	9.997e-01	2.595e-04	-1.187	0.2353
kim1	-4.748e-04	9.995e-01	5.558e-04	-0.854	0.3929
totalba	-4.234e-05	1.000e+00	4.488e-05	-0.944	0.3454
tca	9.139e-04	1.001e+00	5.222e-04	1.750	0.0801 .
totalba_muM	NA	NA	0.000e+00	NA	NA
gcdca	7.997e-05	1.000e+00	1.236e-04	0.647	0.5175
gca	-7.650e-05	9.999e-01	1.177e-04	-0.650	0.5157
gudca	9.459e-05	1.000e+00	6.665e-05	1.419	0.1558

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ngal	0.9955	1.0045	0.9895	1.002
cystatinc	1.0005	0.9995	1.0000	1.001
tcdca	0.9997	1.0003	0.9992	1.000
kim1	0.9995	1.0005	0.9984	1.001
totalba	1.0000	1.0000	0.9999	1.000
tca	1.0009	0.9991	0.9999	1.002
totalba_muM	NA	NA	NA	NA
gcdca	1.0001	0.9999	0.9998	1.000
gca	0.9999	1.0001	0.9997	1.000
gudca	1.0001	0.9999	1.0000	1.000

Concordance= 0.68 (se = 0.05)

Likelihood ratio test= 9.66 on 9 df, p=0.4

Wald test = 9.41 on 9 df, p=0.4

Score (logrank) test = 10.09 on 9 df, p=0.3

Cluster 7 ($n = 46$, events = 27)

=====

Cluster: 7

=====

Call:

```
survival::coxph(formula = f_cox, data = dcl, x = TRUE)
```

```
n= 46, number of events= 27
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
ngal	2.501e-03	1.003e+00	4.490e-03	0.557	0.578
cystatinc	-6.459e-05	9.999e-01	2.864e-04	-0.226	0.822
tcdca	3.107e-05	1.000e+00	3.279e-05	0.948	0.343
kim1	-2.486e-04	9.998e-01	2.402e-04	-1.035	0.301
totalba	-3.216e-05	1.000e+00	3.106e-05	-1.035	0.300
tca	-3.488e-05	1.000e+00	5.946e-05	-0.587	0.557
totalba_muM	NA	NA	0.000e+00	NA	NA
gcdca	4.691e-05	1.000e+00	4.288e-05	1.094	0.274
gca	5.085e-05	1.000e+00	3.225e-05	1.577	0.115
gudca	4.252e-05	1.000e+00	3.143e-05	1.353	0.176

	exp(coef)	exp(-coef)	lower .95	upper .95
ngal	1.0025	0.9975	0.9937	1.011
cystatinc	0.9999	1.0001	0.9994	1.000
tcdca	1.0000	1.0000	1.0000	1.000
kim1	0.9998	1.0002	0.9993	1.000
totalba	1.0000	1.0000	0.9999	1.000
tca	1.0000	1.0000	0.9998	1.000
totalba_muM	NA	NA	NA	NA
gcdca	1.0000	1.0000	1.0000	1.000
gca	1.0001	0.9999	1.0000	1.000
gudca	1.0000	1.0000	1.0000	1.000

```
Concordance= 0.739 (se = 0.051 )
```

```
Likelihood ratio test= 15.82 on 9 df, p=0.07
```

```
Wald test = 18.75 on 9 df, p=0.03
```

```
Score (logrank) test = 35.28 on 9 df, p=5e-05
```

Eidesstattliche Versicherung

(Affidavit)

Seeyakmani Kuson, Sumed

242571

Name, Vorname
(surname, first name)

Matrikelnummer
(student ID number)

☒ Bachelorarbeit
(Bachelor's thesis)

☐ Masterarbeit
(Master's thesis)

Titel
(Title)

Comparison of Survival Prediction Models with LASSO and Random Survival Forests for Patients with Liver Disease

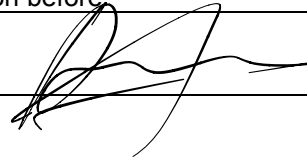
Ich versichere hiermit an Eides statt, dass ich die vorliegende Abschlussarbeit mit dem oben genannten Titel selbstständig und ohne unzulässige fremde Hilfe erbracht habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate kenntlich gemacht. Die Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

I declare in lieu of oath that I have completed the present thesis with the above-mentioned title independently and without any unauthorized assistance. I have not used any other sources or aids than the ones listed and have documented quotations and paraphrases as such. The thesis in its current or similar version has not been submitted to an auditing institution before.

Dortmund, 22.02.2026

Ort, Datum
(place, date)

Unterschrift
(signature)



Belehrung:

Wer vorsätzlich gegen eine die Täuschung über Prüfungsleistungen betreffende Regelung einer Hochschulprüfungsordnung verstößt, handelt ordnungswidrig. Die Ordnungswidrigkeit kann mit einer Geldbuße von bis zu 50.000,00 € geahndet werden. Zuständige Verwaltungsbehörde für die Verfolgung und Ahndung von Ordnungswidrigkeiten ist der Kanzler/die Kanzlerin der Technischen Universität Dortmund. Im Falle eines mehrfachen oder sonstigen schwerwiegenden Täuschungsversuches kann der Prüfling zudem exmatrikuliert werden. (§ 63 Abs. 5 Hochschulgesetz - HG -).

Die Abgabe einer falschen Versicherung an Eides statt wird mit Freiheitsstrafe bis zu 3 Jahren oder mit Geldstrafe bestraft.

Die Technische Universität Dortmund wird ggf. elektronische Vergleichswerkzeuge (wie z.B. die Software „turnitin“) zur Überprüfung von Ordnungswidrigkeiten in Prüfungsverfahren nutzen.

Die oben stehende Belehrung habe ich zur Kenntnis genommen:

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Any person who intentionally breaches any regulation of university examination regulations relating to deception in examination performance is acting improperly. This offense can be punished with a fine of up to EUR 50,000.00. The competent administrative authority for the pursuit and prosecution of offenses of this type is the Chancellor of TU Dortmund University. In the case of multiple or other serious attempts at deception, the examinee can also be unenrolled, Section 63 (5) North Rhine-Westphalia Higher Education Act (*Hochschulgesetz, HG*).

The submission of a false affidavit will be punished with a prison sentence of up to three years or a fine.

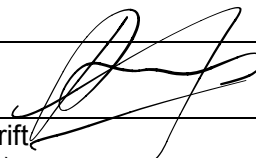
As may be necessary, TU Dortmund University will make use of electronic plagiarism-prevention tools (e.g. the "turnitin" service) in order to monitor violations during the examination procedures.

I have taken note of the above official notification:*

Dortmund, 22.02.2026

Ort, Datum
(place, date)

Unterschrift
(signature)



***Please be aware that solely the German version of the affidavit ("Eidesstattliche Versicherung") for the Bachelor's/ Master's thesis is the official and legally binding version.**

Liste von Positivbeispielen für den Einsatz von IT-/AI-gestützten Sprachmodellen bei der Erstellung von Abschlussarbeiten

Die Verwendung von KI darf nur einen Hilfscharakter haben. Insofern darf KI nur für solche Einsatzzwecke genutzt werden, in denen man in der Lage ist, die Ausgabe der KI zu bewerten. Grundsätzlich dürfen KI-Ergebnisse nicht ungeprüft übernommen werden.

Nachfolgende Liste gibt mögliche Einsatzzwecke von KI an, in welchen KI verwendet werden darf, ohne dies explizit in der eidesstattlichen Versicherung zu vermerken. Jeglicher darüber hinaus gehender Einsatz muss mit den betreuenden Personen abgesprochen und in der eidesstattlichen Versicherung angegeben werden.

Produktname	Art der Verwendung	Ggf. Prompt
z.B. ChatGPT	Brainstorming bei der Forschungsfrage	Was sind mögliche Forschungsfragen zum Thema „X“?
z.B. ChatGPT	Übersicht über das Themengebiet	Fasse mir das Themengebiet / Thema „X“ in maximal 2 Seiten zusammen
z.B. ChatGPT	Hilfe bei der Literatursuche. WICHTIG: Aktuell ist ChatGPT noch nicht in der Lage, verlässlich <i>existierende</i> Paper und Arbeiten aufzuzeigen. Jede gefundene Literatur sollte daher auf Existenz und tatsächlichen Inhalt überprüft werden. Ebenso garantiert ChatGPT keine <i>Vollständigkeit</i> der Literaturrecherche. Deshalb müssen weitere Methoden zur Literaturrecherche (z.B. Suchmaschinen, Vorwärts-/Rückwärtssuche) angewandt werden.	Welche Literatur kannst du mir zu Thema „X“ empfehlen? In welchen Papern oder Büchern kann ich Informationen zu „X“ finden? In welchen Papern wird eine ähnliche Fragestellung wie „X“ untersucht? Mit welchem R-Paket kann ich „X“ machen?
z.B. ChatGPT	Hilfe bei der Programmierung zur Fehlersuche und Präsentation der Ergebnisse (Grafiken, Tabellen, etc.)	Was ist der Fehler im folgenden Code? Wandele mir folgenden base-R-Code in ggplot-Code um. Übersetze meinen Code von R nach Python.
z.B. ChatGPT	Hilfe bei der Formulierung und Verbesserung des Schreibstils	Zeige mir 3 verschiedene Varianten, wie man den folgenden Text alternativ formulieren könnte: „...“
z.B. ChatGPT	Hilfe bei der Ausformulierung von Texten	Wandele meine nachfolgenden Stichpunkte in einen Fließtext um.
z.B. DeepL Write	Optimierung des selbst verfassten Abschnitts XY hinsichtlich Rechtschreibung, Grammatik und Schreibstil	
z.B. DeepL Write	Hilfe bei Übersetzungen und Synonymvorschlägen, vor allem, wenn die Arbeit nicht in der Muttersprache verfasst wird	

List of positive examples how to use IT-/AI-supported language models in the preparation of theses

The use of AI may only be of an auxiliary nature. In this respect, AI may only be used for applications in which you are in a position to evaluate the output of the AI. In principle, AI results must not be used without being checked.

The following list indicates possible uses of AI in which AI may be used without explicitly stating this in the affidavit. Any use beyond this must be agreed with the supervising persons and stated in the affidavit.

Product name	Type of application	exemplary prompt
e.g. ChatGPT	Brainstorming on the research question	What are possible research questions on the topic of "X"?
e.g. ChatGPT	Overview of the topic area	Summarize the subject area / topic "X" in a maximum of 2 pages
e.g. ChatGPT	Help with the literature search. IMPORTANT: ChatGPT is not yet able to reliably list existing papers and works. Any literature found should therefore be checked for existence and actual content. ChatGPT also does not guarantee the completeness of the literature search. Therefore, other methods of literature research (e.g. search engines, forward/backward search) must be used.	What literature can you recommend on the subject of "X"? In which papers or books can I find information on "X"? In which papers is a similar question to "X" investigated? Which R package can I use to do "X"?
e.g. ChatGPT	Help with programming for troubleshooting and presentation of results (graphics, tables, etc.)	What is the error in the following code? Convert the following base R code to ggplot code. Translate my code from R to Python.
e.g. ChatGPT	Help with formulating and improving your writing style	Show me 3 different variants of how the following text could be formulated alternatively: "..."
e.g. ChatGPT	Help with the formulation of texts	Convert my following key points into a continuous text.
e.g. DeepL Write	Optimization of the self-written section XY regarding spelling, grammar and writing style	
e.g. DeepL Write	Help with translations and suggestions for synonyms, especially if the work is not written in the native language	

Ergänzung zur Eidesstattlichen Versicherung (Affidavit)
hinsichtlich des Einsatzes von generativen IT-/KI-gestützten Sprachmodellen

Seeyakmani Kuson, Sumed

242571

Name, Vorname
(surname, first name)

Matrikelnummer
(student ID number)

Comparison of Survival Prediction Models with LASSO and Random Survival Forests for Patients

Titel
(title) ☒ **Bachelorarbeit** ☐ **Masterarbeit**
(bachelor thesis) (master thesis)

In Ergänzung zur abgegebenen eidesstattlichen Versicherung (Affidavit) versichere ich, dass ich bei der Erstellung der vorliegenden Abschlussarbeit betreut von (Erstbetreuer):
Prof. Dr. Jörg Rahnenführer

und (Zweitbetreuer):
Marieke Stolte

im Falle des Einsatzes von generativen IT-/KI-gestützten Sprachmodellen diese Werkzeuge in der Übersicht verwendeter Hilfsmittel mit ihrem Produktnamen und der Art des Hilfsmittels, der Art der Verwendung und ggf. den jeweiligen Prompts vollständig aufgeführt habe (siehe beispielhaft nachfolgende Tabelle). Erlaubte Einsatzzwecke sind unter anderem Optimierung selbstverfasster Texte hinsichtlich Grammatik, Rechtschreibung und Schreibstil, Brainstorming bei der Forschungsfrage oder Unterstützung bei der Literaturrecherche, die die Eigenständigkeit bei der Bearbeitung der Abschlussarbeit nicht einschränken. Ferner versichere ich, dass über die genannten Beispiele hinausgehende Verwendung von IT-/KI-gestützten Sprachmodellen mit dem/der Betreuer:in der Arbeit abgesprochen wurde. Auf die Belehrung hinsichtlich der Folgen von Täuschungsversuchen und der Abgabe einer falschen Versicherung an Eides statt wird auf die eidesstattliche Versicherung und die Prüfungsordnung verwiesen. Diese Belehrung habe ich zur Kenntnis genommen.

In addition to the affidavit submitted, I hereby affirm that I have used generative IT-/AI-supported language models in the preparation of this thesis under the supervision of (first supervisor):
Prof. Dr. Jörg Rahnenführer

and (second supervisor):
Marieke Stolte

and that I have listed these tools in full in the overview of tools used with their product name and the type of tool, the type of use and, if applicable, the respective prompts (see table below as an example). Permitted uses include optimization of self-authored texts with regard to grammar, spelling and writing style, brainstorming on the research question or support in literature research that does not restrict independence in the processing of the thesis. Furthermore, I assure that the use of IT-/AI-supported language models beyond the examples mentioned has been discussed with the supervisor(s) in the thesis. Please refer to the affidavit and the examination regulations for information on the consequences of attempted cheating and making a false declaration in lieu of an oath. I have taken note of these instructions.

Verwendete IT-/KI-gestützte Sprachmodelle IT/AI-supported language models used

Produktname Productname	Art des Hilfsmittels Type of tool	Art der Verwendung Type of use	Prompt	Textstelle(n) Text passage(s)
ChatGPT	KI-Textgenerator/ AI text generator	Help with LaTeX coding, with table formulation	How can i turn this R table output into LaTeX Code?	-
Grammarly	KI- Schreibassistent/	Optimization of chapter 1&5, with spelling, grammar and sentence flow	-	1&5
ChatGPT	KI-Textgenerator/ AI-Textgenerator	Formatting and R code troubleshooting	How do i reduce the whitespace around the rpart.plot?	-

Dortmund, 22.02.2026

Ort, Datum
(place, date)

Unterschrift
(signature)

